

START

1. Start here

UPDATE · 29 SEP 2026 · V3

The architecture review was checked point by point. Most of it was already here, so v3 adds the depth and the missing pieces.

This page summarises the update. Each item links to the section where the detail lives. Tick sections off in the checklist at the bottom as you review them.

SIMPLE VERSION · 1 Free requirements Discover · ₹0 · 5 large-cap picks	SIMPLE VERSION · 2 Plus requirements Decide · full list, sizing, real-time alerts	SIMPLE VERSION · 3 Pro requirements TradeMind · all intelligence, copilot, research
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Check these three first. Everything after them in the sidebar is the deep explanation behind each line.

Added · your request

Bear market memory and playbook

The app remembers every past correction and bear phase: how deep and long it was, what our system did, and what it lost. When a similar phase starts, it shows the closest past episode, switches to a tested defensive playbook, and reviews each user's holdings. Two new items: F35 market memory (Month 1) and F36 bear mode and playbook (Month 2).

Important finding: we don't yet know what the system lost in past bear phases. The replay averaged 8-month windows and skipped the regime capital limit, so measuring this is the first Month 1 task.

[Open Bear market memory →](#)

Latest addition · your request

How it works: flowcharts, algorithms and attach points

A new sidebar group with four sections: a **system map** of the whole application, from data to what each plan sees, with **12 numbered attach points** where intelligence plugs in and a guide for attaching new intelligence later; the **daily workflow** lane by lane with the nightly decision algorithm; the **trade lifecycle** from signal to lesson with the learning algorithms; and the **plan gating flow** that decides what Free, Plus and Pro see. Nine algorithms in plain pseudocode.

[Open System map →](#) [Daily workflow →](#) [Trade lifecycle →](#) [Plan gating →](#)

Added · your request

Swing trading research and competitor notes

Five swing-trading articles and the Tickertape recording and screenshot, checked one by one. **Finding:** none of the sources has a model or evidence, only rules; our app already follows most of their common playbook. Every item is classified as **Have, Add, Test first, Later** or **Not required**. Added: a setup playbook with exact rules for 7 setups, the new building blocks, test list T1-T10, and new items F40 trigger entry, F41 structure exits, F42 big-investor deals, F43 community benchmark, F44 ready-made screens, F45 tax view, F46 Indian theme rotation.

Needs your decision: setup-specific exits would change the locked +8% / -4% / 15-bar spec; only if replay proves it.

Also added: the **Upgrade and checkout page** (F48) from the Tickertape Pro pricing you shared (₹399 a month, ₹899 for 3 months, ₹2,399.2 a year, 20% coupon), with coupons in the admin panel. Longer billing periods depend on SEBI's advance-fee limit. [Open Upgrade and checkout →](#)

The **Sign-up and sign-in pop-up** (F49), from the Tickertape pop-up you shared: an app-install QR panel beside the form; password or Google now, +91 phone OTP at go-live; consent to the RA and AI-use disclosures. [Open Sign-up and sign-in →](#)

The **Stock page** (F47), based on the stock screen you shared: header card with our decision first, a chart with our target, trigger and stop lines, and tabs for every piece of intelligence. It's marked as the first screen to build when you give the go-ahead. [Open Stock page →](#)

[Open Swing trading research →](#) [Open Competitor notes →](#)

Added · your request

Admin panel, deployment and testing

Everyone has their own account; you are the owner. The **admin panel** manages users, invites and subscriptions, including your own. You can change your plan for real, or "view as" Free, Plus or Pro without changing anything. It also holds the plan catalogue, the approvals inbox, system and safety controls, compliance and revenue, and every action is logged. **Sign-in:** username and password or Google for now; OTP login and Okta at go-live. **Deployment:** one product for all plans, your trading core kept apart, a new staging environment, and testing with your own Zerodha. New items: P6 and P7 (Month 4).

[Open Admin panel →](#) [Open Deployment and testing →](#)

Added · your request

Experience and scorecards

Scores, factors and filters are judged by their own track record. The **score report card** shows what each score band has actually delivered, and flags a band that falls short. The **filter and factor scorecard** follows what each filter blocked and what those stocks did afterwards. The **lessons loop** turns every bear phase into a better playbook, tested on past episodes before use. New items: F37, F38 (Month 2), F39 (Month 3).

Honest limit: no score can make returns mandatory, and SEBI bans promising returns. What's mandatory is accountability: every score shows what it has delivered.

[Open Experience and scorecards →](#)

1 · What the review check found

Already in v2 · expanded

Review points 1–21

Meta confidence, Stock DNA, setup models, Expected R, time horizons, news, event shocks, why not buy, what changed, portfolio brain, opportunity cost, failure investigator, model agreement, regime change, trading twin, copilot, morning brief, Monte Carlo, what-if, Research lab, and the non-ML gaps.

Each was a single short card in v2. Each now has its own section with a worked example.

[See the point-by-point table →](#)

Missing in v2 · added now

Frozen architecture diagram	Architecture
Priority order: Core → Differentiation → Moat	6-month roadmap
Advanced portfolio optimisation (F33)	Portfolio brain
Event-impact learning (F34)	News brain
Portfolio, watchlist and broker-connection ownership	Data model
API surface by plan	APIs
12-month moat plan	12-month moat

2 · Where v3 doesn't follow the review

These six changes keep the numbers honest and the product legal. Everything else in the review is adopted.

#	THE REVIEW SAID	V3 DOES INSTEAD	WHY	DETAIL
1	Add news, sector and risk as separate adjustments ("decision impact +7%")	All inputs go into one calibrated meta-model	Stacked adjustments count the same evidence twice, so "70%" stops meaning 70%	Decision engine
2	Show "similar events 82%, typical 10-day move +4.1%" for news	Classify news from Month 5; show impact numbers after about 12 months	Filing history starts only when we start recording it	News brain
3	The user profile feeds Stock DNA	The user profile feeds ranking, portfolio and risk	DNA is a fact about a stock and must be the same for everyone	Architecture

#	THE REVIEW SAID	V3 DOES INSTEAD	WHY	DETAIL
4	P0 is all intelligence features	Same order, but on your paper account first, with the product track alongside	No public calls before SEBI registration, a data licence and paper evidence of profit	6-month roadmap
5	Fundamentals in the Data Brain	Planned for the 12-month moat (NSE XBRL)	There is no point-in-time fundamentals source today	12-month moat
6	Per-stock setup percentages ("Breakout 78%")	Shown only with 20+ past setups in that stock, with the sample size	Too few samples would give confident-looking noise	Stock DNA

3 • One correction to the review's numbers

STOCK	P(WIN)	AVG WIN	REVIEW SAID	CORRECT
A	68%	+2.1R	+1.10R	+1.11R
B	78%	+1.2R	+0.72R	+0.72R
C	61%	+2.8R	+1.11R	+1.32R

EV = P(win) × avg win - P(loss) × avg loss

For C: 0.61 × 2.8 – 0.39 × 1 = +1.32R. The conclusion gets stronger: the lowest-confidence stock has the best edge. [See the Decision engine →](#)

4 • Decisions I need from you

#	DECISION	OPTIONS	MY RECOMMENDATION
1	Freeze the architecture , including the six changes above	Approve · request changes	Approve
2	Priority order	Core → Differentiation → Moat beside the product track · the review's order as is	Core → Differentiation → Moat, beside the product track
3	Go multi-user?	Yes, build a paid product · stay a personal tool	Your call. Yes commits us to SEBI registration, a data licence and billing
4	Month 1 budget	Consultant, data-vendor quotes, NISM exam	Collect quotes now; no spending commitment yet
5	Language model	Claude API · none	Claude API, for the brief's wording, news classification and the copilot

Reply in chat with your answers or changes. After approval I write the Month 1 implementation plan, you choose how it runs, and building starts behind feature flags.

5 • Review checklist

Tick each section once you've read it. Saved in this browser only.

0 of 40 sections reviewed

[Clear ticks](#)

- | | | | | |
|--|---|---|--|---|
| ☐ Free requirements · Plans · simple version | ☐ Plus requirements · Plans · simple version | ☐ Pro requirements · Plans · simple version | ☐ System map and attach points · How it works | ☐ Daily workflow and decision algorithm · How it works |
| ☐ Trade lifecycle and learning · How it works | ☐ Plan gating flow · How it works | ☐ Overview · Background | ☐ Review check · Background | ☐ Current system · Background |
| ☐ Swing trading research · Research | ☐ Competitor notes: Tickertape · Research | ☐ Market brain · Deep dive · intelligence | ☐ Bear market memory · Deep dive · intelligence | ☐ Stock DNA · Deep dive · intelligence |
| ☐ ML models · Deep dive · intelligence | ☐ News brain · Deep dive · intelligence | ☐ Decision engine · Deep dive · intelligence | ☐ Portfolio brain · Deep dive · intelligence | ☐ AI copilot and brief · Deep dive · intelligence |
| ☐ Learning and failure brain · Deep dive · intelligence | ☐ Experience and scorecards · Deep dive · intelligence | ☐ Research lab · Deep dive · intelligence | ☐ Plans and monetization · Deep dive · product | ☐ User controls · Deep dive · product |
| ☐ Stock page · Deep dive · product | ☐ Sign-up and sign-in · Deep dive · product | ☐ Upgrade and checkout · Deep dive · product | ☐ Admin panel · Deep dive · product | ☐ Subscriptions and costs · Deep dive · product |
| ☐ Not possible · Deep dive · product | ☐ Architecture · Deep dive · engineering | ☐ Data model · Deep dive · engineering | ☐ APIs · Deep dive · engineering | ☐ Deployment and testing · Deep dive · engineering |
| ☐ Daily data · Deep dive · engineering | ☐ Gap register · Roadmap and decisions | ☐ 6-month roadmap · Roadmap and decisions | ☐ 12-month moat · Roadmap and decisions | ☐ Decisions and next steps · Roadmap and decisions |

FREE · DISCOVER

See what the models think, safely, and learn

For new investors with ₹50,000 to ₹3 lakh who trade a few times a month.

₹0

Price

5 a day

Large-cap picks

End of day

Stop-loss alerts

Month 6

Public launch, if gates pass

Plan basics

AREA	FREE GETS	ACCEPTED WHEN
Recommendations	Top 5 a day, large cap only, from the base safety model and the large-cap safety strategy. Each shows BUY / SELL / HOLD, entry, target, stop and a HIGH / MEDIUM / LOW label	Never more than 5; all large cap
When picks appear	After the 16:00 scan, in the app	Published by 16:15 on trading days
Stop-loss alerts	SL HIT at end of day, in the app	Never delivered in real time
Market	Regime label (risk-on, neutral, risk-off) and the bear-mode banner	Same label as the paid plans on the same day
Morning brief	Short version at 08:45	Delivered by 08:45 on trading days
Why not buy	The top reason for any stock	Every non-BUY stock shows one reason
Watchlist	Up to 5 stocks	Sixth stock prompts an upgrade
Trade history and performance	Last 30 days; weekly and monthly summary	Performance numbers are PaRRVA-verified only
Paper trading	Virtual money on the same pipeline as the bot's paper broker	Kept separate from any real-money view
Learning	Model details, strategy catalogue (read-only), guides on stop-loss and sizing	—
Upgrade teasers	Calculator demo on a fixed ₹1 lakh; scan count only ("42 stocks passed today")	No Plus or Pro data leaks through a teaser
Channels	In app; push for the brief; weekly email	—

AREA	FREE GETS	ACCEPTED WHEN
Compliance	RA disclaimer and AI-use note on every screen and alert	Audit row for every recommendation

Intelligence features in Free

FEATURE	THIS PLAN GETS	STATUS	READY	DETAILS
Regime and deployable capital ^{A1}	Regime label	A · Built	Built	Open
Landmine filter ^{A3}	Applied to every pick	A · Built	Built	Open
Exit policy ^{A6}	SL HIT at end of day	A · Built	Built	Open
Market intelligence layer ^{F01}	Risk-on / Neutral / Risk-off	B · Improve	Month 1 · Oct	Open
Meta confidence system ^{F04}	HIGH / MEDIUM / LOW	B · Improve	Month 3 · Dec	Open
News classification ^{F08}	Event badges	C · Build new	Month 5 · Feb	Open
“Why not buy?” engine ^{F10}	Top reason	B · Improve	Month 1 · Oct	Open
Exit intelligence ^{F13}	SL HIT (end of day)	B · Improve	Month 5 · Feb	Open
Trading journal ^{F18}	Paper trades	B · Improve	Month 1 · Oct	Open
Morning brief ^{F21}	Short	C · Build new	Month 2 · Nov	Open
Explainability ^{F22}	Top 3 reasons	B · Improve	Month 3 · Dec	Open
Data quality ^{F24}	Badge	B · Improve	Month 1 · Oct	Open
Uncertainty and model agreement ^{F25}	“Low confidence: wait”	C · Build new	Month 3 · Dec	Open
Strategy lab ^{F27}	Read-only catalogue	B · Improve	Month 6 · Mar	Open
Market memory (episode library) ^{F35}	Closest episode, one line	C · Build new	Month 1 · Oct	Open
Bear mode and playbook ^{F36}	Bear-mode banner	C · Build new	Month 2 · Nov	Open
Score report card ^{F37}	What each label has delivered	B · Improve	Month 2 · Nov	Open
Big-investor deals ^{F42}	Top deals of the day	B · Improve	Month 3 · Dec	—
Ready-made screens ^{F44}	2 screens	C · Build new	Month 4 · Jan	—
Tax view ^{F45}	Estimated tax on paper trades	C · Build new	Month 4 · Jan	—

FEATURE	THIS PLAN GETS	STATUS	READY	DETAILS
Theme rotation F46	Top 3 trending themes	C · Build new	Month 3 · Dec	—
Stock page F47	Header, decision label, daily chart, top reason	B · Improve	Month 1 · Oct	—

Not in Free

- Risk engine and capital ladder · Plus
- Calibration, walk-forward, Model Lab · Plus
- Sector brain · Plus
- Stock DNA · Plus
- Time-horizon probabilities · Pro
- Expected value engine · Plus
- Setup-specific models · Plus
- Event shock detector · Pro
- “What changed?” engine · Plus
- Position intelligence · Plus
- Opportunity cost engine · Pro
- Portfolio health · Plus
- Correlation engine · Pro
- Factor exposure · Pro
- Personal trading twin · Pro
- AI copilot · Pro
- Model observatory · Plus
- Regime change detector · Plus
- Research lab · Pro
- Monte Carlo stress testing · Pro
- What-if simulator · Pro
- Failure investigator · Pro
- Advanced portfolio optimisation · Pro
- Event-impact learning · Pro
- Filter and factor scorecard · Pro
- Lessons loop · Plus
- Trigger entry · Plus
- Structure stops, targets and time stop · Plus
- Community diversification benchmark · Plus

PLUS · DECIDE

Every signal, with the reasons and the size for your money

For part-time swing traders with ₹3 lakh to ₹15 lakh who hold for 1 to 3 weeks.

≈ half of Pro

Price

Real time

Stop-loss alerts

Full list

All caps

Month 6

Public launch, if gates pass

Plan basics · everything in Free, plus

AREA	PLUS GETS	ACCEPTED WHEN
Recommendations	Full list, all caps, with 5 sub-scores, Expected R, and evidence for and against	Same decisions as Pro on the same day
Alerts	SL HIT and exit states in real time, by push and Telegram	Delivered within 2 minutes of the trigger
Market	Index trends, sector arrows, regime-change alerts, bear-mode card with the closest past episode	Regime change alert on the day it's confirmed
Morning brief	Full version	Delivered by 08:45
Models and strategies	Choose 1 model and 1 strategy	Filters the shared scan; a second choice prompts an upgrade
Capital and sizing	Enter capital; deployed vs free; suggested quantity and ₹ amount; risk fixed at 1% per trade	Sizes never exceed the locked limits
Diversification calculator	Any capital, with presets (Conservative, Balanced, Aggressive)	Result labelled as a plan from the user's own inputs
Holdings	Link a broker read-only or upload a file; HOLD, MONITOR, REDUCE or EXIT for each holding	No order is ever placed for the user
Portfolio	Health score	Score explained by its parts
What changed	For the watchlist, every evening	Ready by 16:15
Scan results	Full end-of-day scan with cap and sector filters	—
Watchlist and history	25 stocks; 1 year of trade history	—
Score report card	What each score band has delivered	Bands under 30 signals show “building record”
Channels	In app, push, daily email, Telegram	Quiet hours respected

Intelligence features in Plus

FEATURE	THIS PLAN GETS	VS FREE	STATUS	READY	DETAILS
Regime and deployable capital ^{A1}	Regime + deployable %	Upgraded	A · Built	Built	Open
Risk engine and capital ladder ^{A2}	Fixed 1% sizing	New	A · Built	Built	Open
Landmine filter ^{A3}	Applied, with reasons	Upgraded	A · Built	Built	Open
Calibration, walk-forward, Model Lab ^{A5}	Summary	New	A · Built	Built	Open
Exit policy ^{A6}	Real time	Upgraded	A · Built	Built	Open
Market intelligence layer ^{F01}	+ NIFTY, BANKNIFTY, MIDCAP, SMALLCAP trends	Upgraded	B · Improve	Month 1 · Oct	Open
Sector brain ^{F02}	Sector ↑ → ↓	New	B · Improve	Month 2 · Nov	Open
Stock DNA ^{F03}	4-trait summary	New	C · Build new	Month 4 · Jan	Open
Meta confidence system ^{F04}	5 sub-scores + Expected R	Upgraded	B · Improve	Month 3 · Dec	Open
Expected value engine ^{F06}	Expected R per pick	New	C · Build new	Month 2 · Nov	Open
Setup-specific models ^{F07}	Choose setups	New	C · Build new	Month 2 · Nov	Open
News classification ^{F08}	Event list	Upgraded	C · Build new	Month 5 · Feb	Open
“Why not buy?” engine ^{F10}	Why and why not	Upgraded	B · Improve	Month 1 · Oct	Open
“What changed?” engine ^{F11}	Watchlist	New	B · Improve	Month 3 · Dec	Open
Position intelligence ^{F12}	Action state	New	B · Improve	Month 4 · Jan	Open
Exit intelligence ^{F13}	HOLD / REDUCE / TRAIL / EXIT	Upgraded	B · Improve	Month 5 · Feb	Open
Portfolio health ^{F15}	Health score	New	B · Improve	Month 4 · Jan	Open
Trading journal ^{F18}	Trades + outcomes	Upgraded	B · Improve	Month 1 · Oct	Open
Morning brief ^{F21}	Full	Upgraded	C · Build new	Month 2 · Nov	Open
Explainability ^{F22}	For and against	Upgraded	B · Improve	Month 3 · Dec	Open
Model observatory ^{F23}	Summary	New	B · Improve	Month 6 · Mar	Open
Data quality ^{F24}	Badge	Same	B · Improve	Month 1 · Oct	Open

FEATURE	THIS PLAN GETS	VS FREE	STATUS	READY	DETAILS
Uncertainty and model agreement F25	+ reasons	Upgraded	C · Build new	Month 3 · Dec	Open
Regime change detector F26	Alert	New	B · Improve	Month 1 · Oct	Open
Strategy lab F27	Choose 1	Upgraded	B · Improve	Month 6 · Mar	Open
Market memory (episode library) F35	Episode card: what happened, what we did	Upgraded	C · Build new	Month 1 · Oct	Open
Bear mode and playbook F36	+ holdings flags, why fewer picks	Upgraded	C · Build new	Month 2 · Nov	Open
Score report card F37	Report card by score band	Upgraded	B · Improve	Month 2 · Nov	Open
Lessons loop F39	Lesson being applied	New	C · Build new	Month 3 · Dec	Open
Trigger entry F40	Trigger price on each pick	New	C · Build new	Month 1 · Oct	—
Structure stops, targets and time stop F41	Stop and target levels	New	C · Build new	Month 2 · Nov	—
Big-investor deals F42	All deals + your watchlist	Upgraded	B · Improve	Month 3 · Dec	—
Community diversification benchmark F43	Your score vs community	New	C · Build new	Month 6 · Mar	—
Ready-made screens F44	All ready screens	Upgraded	C · Build new	Month 4 · Jan	—
Tax view F45	On your trades	Upgraded	C · Build new	Month 4 · Jan	—
Theme rotation F46	Theme ranks + their stocks	Upgraded	C · Build new	Month 3 · Dec	—
Stock page F47	+ Expected R, setup, trigger, alerts, intraday chart, full evidence	Upgraded	B · Improve	Month 1 · Oct	—

Not in Plus (Pro only)

- Time-horizon probabilities · Pro
- Event shock detector · Pro
- Opportunity cost engine · Pro
- Correlation engine · Pro
- Factor exposure · Pro
- Personal trading twin · Pro
- AI copilot · Pro
- Research lab · Pro
- Monte Carlo stress testing · Pro
- What-if simulator · Pro
- Failure investigator · Pro
- Advanced portfolio optimisation · Pro
- Event-impact learning · Pro
- Filter and factor scorecard · Pro

PRO · TRADEMIND

Understands the market, your portfolio and its own track record

For active traders with ₹15 lakh or more who want control over models, strategies and risk.

50% below market

Price

6 months free

At launch, if SEBI allows

All channels

Including WhatsApp

Months 2–12

Features arrive in stages

Plan basics · everything in Plus, plus

AREA	PRO GETS	ACCEPTED WHEN
Models and strategies	Any number; agreement mode (BUY only when models agree); tune settings; saved presets	Untested combinations labelled “untested” and never traded with real money
Risk settings	0.25% to 3% risk per trade; maximum daily loss; maximum open positions; exposure per stock and sector	Every setting stays inside the locked maximums
Diversification calculator	Fully flexible; what-if mode; order-list export for the broker	Export matches the plan shown
Scan results	Intraday refresh; all filters	—
Alerts	All channels, including WhatsApp; event-shock alerts	Shock alert within 2 minutes
Watchlist and history	Unlimited watchlist; full history; CSV and PDF export	—
Deep intelligence	Stock DNA, setup models, time horizons, Expected R ranking, news brain, exit intelligence, portfolio brain, opportunity cost, full bear-market memory, scorecards	Each feature passes its own gate before release (see the table below)
AI copilot	Cited answers over our data, with a daily question cap	≥ 90% correct on the evaluation set; no personal order instructions
Research lab	Backtests, then walk-forward, Monte Carlo and what-if, within a job quota	Lab result equals the CLI result

Intelligence features in Pro

FEATURE	THIS PLAN GETS	VS PLUS	STATUS	READY	DETAILS
Regime and deployable capital ^{A1}	Full rule view	Upgraded	A · Built	Built	Open

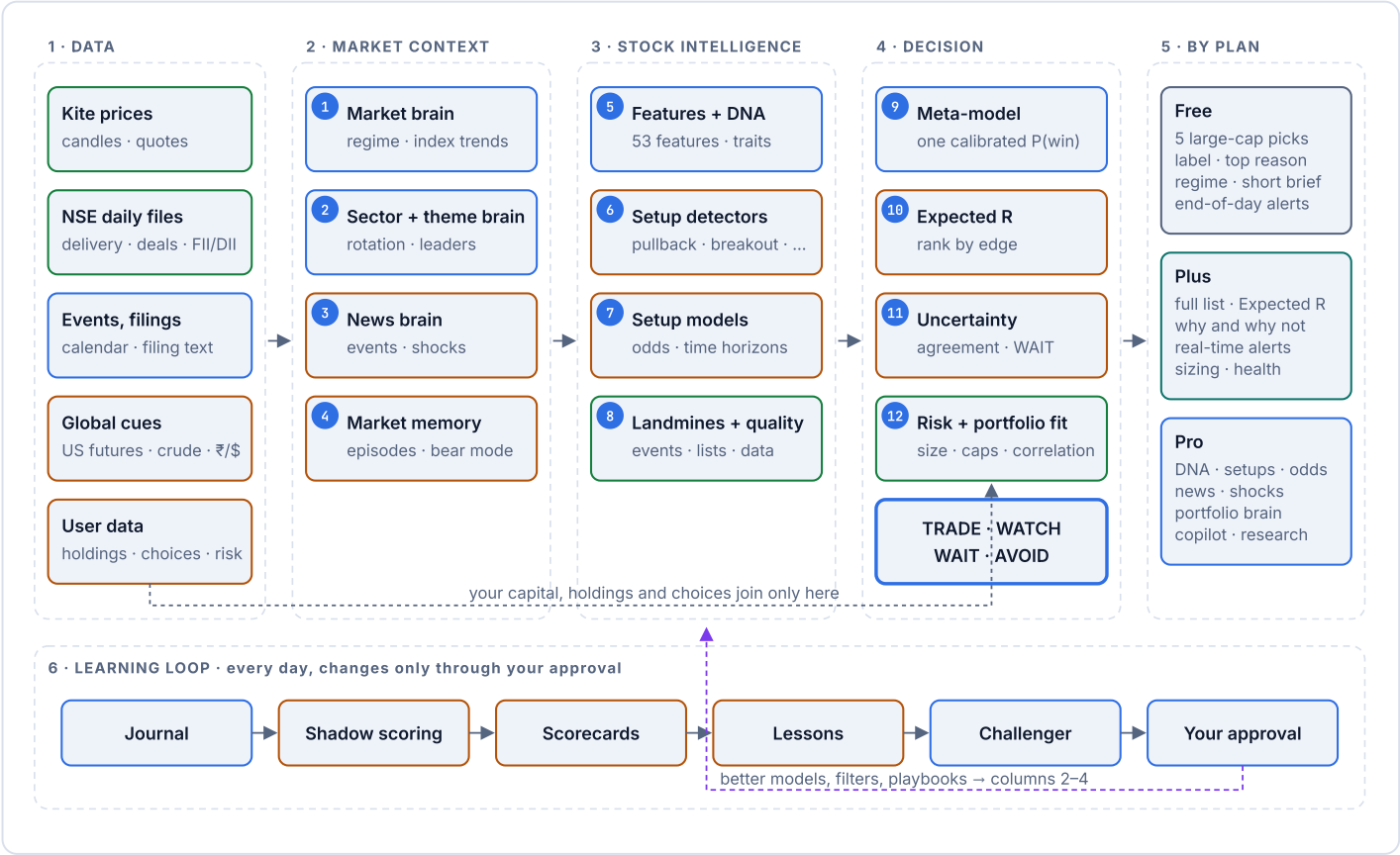
FEATURE	THIS PLAN GETS	VS PLUS	STATUS	READY	DETAILS
Risk engine and capital ladder ^{A2}	Custom limits	Upgraded	A · Built	Built	Open
Landmine filter ^{A3}	Applied, with reasons	Same	A · Built	Built	Open
Calibration, walk-forward, Model Lab ^{A5}	Full	Upgraded	A · Built	Built	Open
Exit policy ^{A6}	Real time + trailing	Upgraded	A · Built	Built	Open
Market intelligence layer ^{F01}	+ global cues, liquidity	Upgraded	B · Improve	Month 1 · Oct	Open
Sector brain ^{F02}	+ industry, leaders, deterioration	Upgraded	B · Improve	Month 2 · Nov	Open
Stock DNA ^{F03}	Full profile, setups, regime, holding, failures	Upgraded	C · Build new	Month 4 · Jan	Open
Meta confidence system ^{F04}	+ calibration and model detail	Upgraded	B · Improve	Month 3 · Dec	Open
Time-horizon probabilities ^{F05}	3, 5, 10, 20-day odds + stop first	New	B · Improve	Month 4 · Jan	Open
Expected value engine ^{F06}	Ranking by Expected R	Upgraded	C · Build new	Month 2 · Nov	Open
Setup-specific models ^{F07}	Per-setup probabilities	Upgraded	C · Build new	Month 2 · Nov	Open
News classification ^{F08}	+ materiality, sentiment, horizon	Upgraded	C · Build new	Month 5 · Feb	Open
Event shock detector ^{F09}	Real-time shock alerts	New	C · Build new	Month 5 · Feb	Open
“Why not buy?” engine ^{F10}	+ what would change it	Upgraded	B · Improve	Month 1 · Oct	Open
“What changed?” engine ^{F11}	All + portfolio	Upgraded	B · Improve	Month 3 · Dec	Open
Position intelligence ^{F12}	+ thesis tracking	Upgraded	B · Improve	Month 4 · Jan	Open
Exit intelligence ^{F13}	+ exit reasons and score	Upgraded	B · Improve	Month 5 · Feb	Open
Opportunity cost engine ^{F14}	Held vs new edge	New	C · Build new	Month 6 · Mar	Open
Portfolio health ^{F15}	Six scores + clusters	Upgraded	B · Improve	Month 4 · Jan	Open
Correlation engine ^{F16}	Clusters, beta, cap concentration	New	C · Build new	Month 2 · Nov	Open
Factor exposure ^{F17}	Price factors now; fundamental later	New	B · Improve	Month 4 · Jan	Open
Trading journal ^{F18}	Full context + failure labels	Upgraded	B · Improve	Month 1 · Oct	Open
Personal trading twin ^{F19}	Personal ranking	New	D · Future	Months 7–12	Open

FEATURE	THIS PLAN GETS	VS PLUS	STATUS	READY	DETAILS
AI copilot F20	Cited answers over your data	New	D · Future	Month 6 · Mar	Open
Morning brief F21	+ your portfolio	Upgraded	C · Build new	Month 2 · Nov	Open
Explainability F22	+ feature contributions	Upgraded	B · Improve	Month 3 · Dec	Open
Model observatory F23	Full lab	Upgraded	B · Improve	Month 6 · Mar	Open
Data quality F24	Per-source detail	Upgraded	B · Improve	Month 1 · Oct	Open
Uncertainty and model agreement F25	+ agreement count	Upgraded	C · Build new	Month 3 · Dec	Open
Regime change detector F26	+ re-rank and re-risk	Upgraded	B · Improve	Month 1 · Oct	Open
Strategy lab F27	All + stats	Upgraded	B · Improve	Month 6 · Mar	Open
Research lab F28	Backtest v1; walk-forward and Monte Carlo later	New	B · Improve	Month 6 · Mar	Open
Monte Carlo stress testing F29	Drawdown range, survival	New	C · Build new	Months 7–12	Open
What-if simulator F30	NIFTY, VIX, cap, sector shocks	New	C · Build new	Months 7–12	Open
Failure investigator F31	Loss labels and totals	New	C · Build new	Month 5 · Feb	Open
Advanced portfolio optimisation F33	Suggested weights to accept or edit	New	D · Future	Months 7–12	Open
Event-impact learning F34	Similar events and typical move	New	D · Future	Months 7–12	Open
Market memory (episode library) F35	Full memory + defence ablations	Upgraded	C · Build new	Month 1 · Oct	Open
Bear mode and playbook F36	+ phase playbook, episode repeat on your portfolio	Upgraded	C · Build new	Month 2 · Nov	Open
Score report card F37	+ by regime and calibration detail	Upgraded	B · Improve	Month 2 · Nov	Open
Filter and factor scorecard F38	Which filters and factors are working, by regime	New	B · Improve	Month 2 · Nov	Open
Lessons loop F39	+ full lessons report per episode	Upgraded	C · Build new	Month 3 · Dec	Open
Trigger entry F40	+ trigger status alerts	Upgraded	C · Build new	Month 1 · Oct	—
Structure stops, targets and time stop F41	+ breakeven and time-stop status	Upgraded	C · Build new	Month 2 · Nov	—
Big-investor deals F42	+ alerts when a deal hits your holdings	Upgraded	B · Improve	Month 3 · Dec	—
Community diversification benchmark F43	+ by capital band	Upgraded	C · Build new	Month 6 · Mar	—

FEATURE	THIS PLAN GETS	VS PLUS	STATUS	READY	DETAILS
Ready-made screens F44	+ custom screens	Upgraded	C · Build new	Month 4 · Jan	—
Tax view F45	+ yearly tax report export	Upgraded	C · Build new	Month 4 · Jan	—
Theme rotation F46	+ theme momentum alerts and theme filter on setups	Upgraded	C · Build new	Month 3 · Dec	—
Stock page F47	+ DNA, setups and odds, peers, overlays, full record	Upgraded	B · Improve	Month 1 · Oct	—

The whole system on one page

Data flows left to right: sources → market context → stock intelligence → one decision → what each plan sees. Everything in columns 2–4 is computed once a day for everyone. The learning loop underneath feeds improvements back, only through your approval. Numbered circles are the **attach points** where intelligence plugs in.



The 12 attach points

#	ATTACH POINT	TAKES IN	GIVES OUT	AFFECTS	ITEMS	READY
1	Market brain	Indices, VIX, breadth, FII/DII, global cues	Regime, deployable %, index trends	How much capital; selection strictness	A1, F01, F26	Built → Month 1
2	Sector and theme brain	Sector and thematic indices, delivery	Ranks, leaders, deterioration	Which stocks are considered first	F02, F46	Months 2–3
3	News brain	NSE filings, price and volume jumps	Event type, materiality, sentiment; shocks	Meta-model input; WAIT near events; alerts	F08, F09, F34	Month 5 → later
4	Market memory	Past episodes, today's market state	Bear-mode phase, closest episode, playbook	Selection and deployable levers	F35, F36, F39	Months 1–3
5	Features and DNA	Candles, flows, events, journal	53 features, traits, setup affinity	Inputs to every model	F03, F17	Month 4
6	Setup detectors	Price structure, EMAs, patterns, volume	Which setup a stock is in, trigger level	No setup = no signal	F07, F40	Month 2
7	Setup models	Features for that setup	Probability and time-horizon odds	Meta-model input	F05, F07	Months 2–4
8	Landmines and data quality	Event calendar, restriction lists, feed freshness	AVOID or WAIT with a reason; shadow record	Blocks entries	A3, F24, F38	Built → Month 2
9	Meta-model	All sub-scores	One calibrated P(win)	Confidence label	F04, F37	Month 3
10	Expected R	P(win), win and loss sizes, costs	Edge in R; ranking	Which TRADE comes first	F06	Month 2
11	Uncertainty	Model votes, data quality, regime change, events	Agreement count; WAIT	Stops forced BUYS	F25	Month 3
12	Risk and portfolio fit	Decision + user capital, holdings, limits	Size, caps, correlation warnings, opportunity cost	Quantity; whether it fits this portfolio	A2, F14–F16, F33	Built → Month 4

How to attach a new intelligence system later

Any future idea (options positioning, fundamentals, social sentiment, a new model) plugs in through one of four doors, and the same rules apply every time.

As context (attach 1–4)

A new market-wide signal, such as option put/call ratio. It becomes a sub-score for the meta-model, never a direct BUY or SELL.

As a filter (attach 8)

A new “stay out” rule. It's shadow-scored from day one, and its verdict on the filter scorecard decides if it stays.

As a setup (attach 6–7)

A new pattern with its own detector and model. It's kept only if it beats “score everything” on the paper stream.

As an explanation (attach 9, 11)

New evidence for or against, shown to users, such as a copilot tool. It changes what users understand, not the decision.

Every door has the same gate: walk-forward and paper evidence, a scorecard with a minimum sample, and your approval in the admin panel's approvals inbox. The locked risk rules (stop, risk per trade, sector cap, drawdown halt) can't be changed through any door.

One trading day, lane by lane

TIME (IST)	TRADING CORE (PRIVATE)	PRODUCT STACK	FREE	PLUS	PRO
06:10–06:30	Kite login; fetch global cues, FII/DII, events, restriction lists	—	—		
08:30	Market brain [1]; regime-change check; bear-mode phase [4]	Receives market state	—		
08:45	—	Builds briefs per plan and per user	Short brief	Full brief, push	+ your holdings, all channels
09:15–15:30	Quotes every 60 s; stops, targets, trailing; trigger entries [6]; shock detector [3]	Alerts routed by plan	Nothing until evening	Real-time stop and exit alerts	+ shock alerts
15:40	Daily candles, bhavcopy, delivery, deals	—	—		
15:45	Nightly decision algorithm (below), attach points 1–12	—	—		
16:00	Publishes decisions one way	Stores and gates by plan	5 large-cap picks	Full list	+ EV ranking, odds
16:15	Signal scoring, shadow scoring, scorecards, what changed	Position intelligence per user [12]	End-of-day stop alerts	Watchlist changes, holding states	+ portfolio brain, opportunity cost
Evening	Backups; monthly challenger training	Copilot, research jobs, admin approvals	—	—	Copilot, research lab

Algorithm 1 · Nightly decision pipeline (15:45)

```
every trading day after the daily candles load:
    market = market_brain.today()                # regime, trends, breadth, VIX    [1]
    if regime has changed for 2 days in a row:
        regime_change()                            # re-score, re-rank, re-risk, notify
    phase = bear_mode.phase(market, market_memory) # normal · early · decline · panic · recovery    [4]
    sectors = sector_theme_brain.rank(today)        # sectors and themes            [2]
    universe = active NSE stocks passing liquidity and price floors

    for each stock in universe:
        if data_quality(stock) is below threshold:
            save decision(stock, WAIT, reason = "data incomplete"); continue    [8]
        if landmine(stock):
            # results soon, ASM/GSM/T2T, corporate action
            save decision(stock, AVOID, reason); shadow_score(stock); continue    [8]
        features = build_features(stock); dna = stock_dna(stock)                [5]
        setups = detect_setups(stock, features) # pullback, breakout, momentum, ... [6]
        if no setups: save decision(stock, NO SETUP); continue
```

```

for each setup s:
    p_setup = setup_model[s].predict(features)    # + time-horizon heads [7]
    p_win   = meta_model(rule_score, p_setup, market, sector, news, risk_quality) [9]
    exp_r   = p_win × avg_win(s, regime) - (1 - p_win) × avg_loss(s, regime) - costs [10]
    agree   = models voting BUY ÷ models voting [11]
    state   = decide(p_win, exp_r, agree, phase, entry_zone)
    save decision(stock, s, p_win, exp_r, agree, state,
                  evidence_for, evidence_against, trigger = signal_high × 1.0025)

ranked = TRADE decisions ordered by exp_r, highest first
for each book (bot paper/live, and each user's view):
    fit = portfolio_fit(ranked, holdings, correlation, sector cap, deployable %) [12]
publish decisions to the product stack (one way)

```

Algorithm 2 · Deciding the state

```

decide(p_win, exp_r, agree, phase, entry_zone):
    if exp_r ≤ 0 or risk quality is poor:    return AVOID
    if agree below 4 of 6, or phase is panic,
        or an event is due, or regime changing: return WAIT
    if price is 3%+ above the entry zone:    return WATCH "wait for pullback"
    if p_win ≥ min_conf(phase)
        and exp_r ≥ min_edge:                return TRADE
    otherwise:                               return WATCH

min_conf: 0.60, plus 0.10 in a NIFTY downtrend (today's rule)
min_edge, 4-of-6 and 3%: starting values, tuned by walk-forward

```

Algorithm 3 · Sizing (today's rule, kept)

```

risk_rupees = capital × risk_per_trade    # 1%, locked
qty_by_risk = floor(risk_rupees ÷ (entry - stop))
qty_by_alloc = floor(allocation ÷ entry)
qty = smallest of qty_by_risk, qty_by_alloc,
        single-stock cap, cash above the cash floor
if deployable % is used up, or drawdown ≥ 15%:
    qty = 0                                # no new entries; exits continue

```

Algorithm 4 · Trigger entry (test T1)

```

next session after a TRADE decision:
    if price trades above trigger and volume pace ≥ 1.5×:
        buy with a limit just above trigger
    else at the close:
        trigger expires; record EXPIRED (still scored)

```

From signal to lesson



Algorithm 7 · Filter verdict from shadow scoring

```

for each filter and regime:
    blocked = resolved shadow signals the filter blocked
    if fewer than 50:         verdict = building record
    elif mean R of blocked ≤ -0.1R: verdict = protecting
    elif mean R of blocked ≥ +0.1R: verdict = costing us
    else:                     verdict = neutral
    keep / tune / drop proposal → approvals inbox
    (locked safety rules are never removed automatically)
  
```

Algorithm 5 · Position and exit loop

```

every 60 s while the market is open:
    if price ≤ stop:           exit (stop)
    if price ≥ first target:   sell half; trail the rest
    if trailing stop is hit:   exit (trail)

every evening for each open position:
    health = meta_model(today's sub-scores) vs at entry
    state = HOLD · MONITOR · REDUCE · EXIT
    change the state only after 2 days in a row (hysteresis)
    if held past the time stop with no progress: exit (time)
    if a better setup's edge beats the remaining edge by 1R+:
        show "review A vs B" (never sell automatically)
  
```

Algorithm 6 · Score band health (daily)

```

for each score band (90+, 80-89, 70-79, 60-69) and regime:
    recent = last 60 resolved signals in the band
    if fewer than 30:           status = building record
    elif stated rate - real hit rate > 10 pts:
        status = under review; band stops showing as BUY
    else:
        status = delivering
  
```

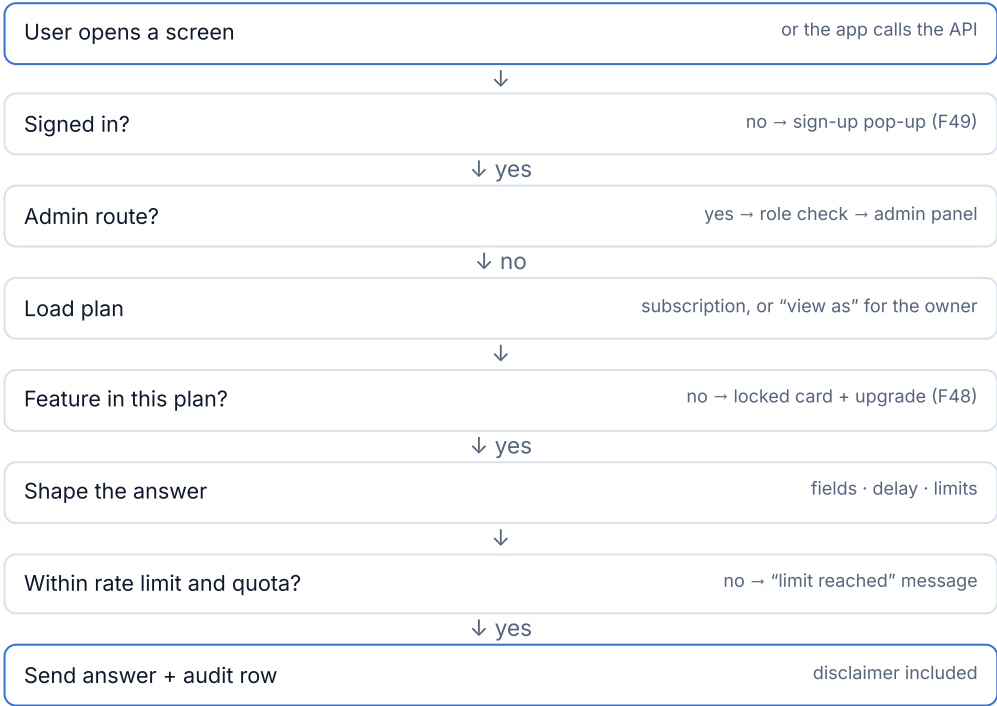
Algorithm 8 · Lessons and promotion

```

when a weak phase ends (regime normal for 2 days):
    report = lessons(episode) # losses, why, what saved us
    for each proposal in report:
        test on the other stored episodes (out of sample)
        test on normal-market periods
        if better out of sample and no worse normally:
            send to the approvals inbox
    on your approval:
        new version runs as a shadow first
        goes live after 30+ resolved signals beat the champion
  
```

How every request decides what Free, Plus and Pro see

All plans call the same API. One gate in `api/deps.py` decides, for each request, whether the user may see a feature and how much of it.



Algorithm 9 · Gate per request

```
user = authenticate(token)
if no user:
    return sign_up_popup()
if route is /admin/*:
    require role in (owner, admin, support)
plan = user.view_as or active_subscription(user).plan
rules = plan_catalog[plan] # editable in the admin panel
if feature not in rules:
    return locked(feature,
                    upgrade_to = cheapest plan that has it)
data = shape(data, rules) # e.g. Free: 5 large-cap picks,
                          # end-of-day alerts, label only
if over rules.rate_limit or rules.quota:
    return limit_reached()
audit(user, feature, plan)
return data + disclaimer
```

SHAPE RULE	FREE	PLUS	PRO
Picks	5, large cap	All	All + EV sort
Alert delay	End of day	Real time	Real time
Decision fields	Label, top reason	+ sub-scores, Expected R	+ odds, agreement
Quotas	Watchlist 5	Watchlist 25	Copilot and research caps

WHAT THIS DOCUMENT IS

SwingTradeML grows from a stock-scoring signal generator into a market decision-intelligence platform.

It reads the market, then the sector, then the stock. It decides TRADE, WATCH, WAIT or AVOID with evidence for and against, and it learns from every outcome. This is the single document for the product, the AI and the build plan.

Keep the existing engine → expose what it already does → fill the missing intelligence layers → measure every new model against today's baseline.

8

A · Built · surface it

24

B · Exists · improve

27

C · Missing · build new

5

D · Future · moat

64

Items in the gap register · 32 build on existing code

What changed since v2

Review checked point by point

All 23 points in the architecture review are answered in **Review check**. 21 were already in v2 as feature cards; v3 gives each a full section with a worked example.

Architecture frozen

Data Brain and User Brain feed one decision pipeline that ends in a journal and a learning loop. It's adopted from the review with one correction (see **Architecture**).

Priorities re-ordered

Core intelligence first, then differentiation, then the moat. This runs in parallel with the product track (accounts, billing, compliance, data licence).

New deep-dive sections

Market brain, Stock DNA, ML models, News brain, Decision engine, Portfolio brain, AI copilot, Learning brain and Research lab.

Engineering detail

Data model with user, portfolio, watchlist and broker-connection ownership, plus the API surface by plan.

Time plan

A 6-month roadmap (Oct 2026 to Mar 2027) with a gate each month, then a 12-month moat plan to Sep 2027.

Three gates still come first

GATE 1 • EDGE

Signals must make money after costs

The replay of the real system on ₹10 lakh returned -0.1% to -0.4% per 8-month window after costs, while NIFTY returned $+6\%$. AUC is 0.556–0.583. The core intelligence track (setup models, EV ranking) is also the most promising route to an edge, so it starts on the owner's paper book.

GATE 2 • MULTI-USER

Separate accounts before paid plans

Trading tables have no per-user owner today. Accounts, portfolios, watchlists, broker connections and plan checks come before any public user.

GATE 3 • LICENCES

SEBI RA and licensed data

Live calls to the public need SEBI Research Analyst registration. Prices shown to users need a licensed data vendor. Public performance needs PaRRVA verification.

REVIEW CHECK

Every point in the review, against v2

Most points were already in v2 as feature cards, so v3 mainly adds depth. Two points were new: the frozen architecture and the priority order. One new feature (advanced portfolio optimisation) and one split (event-impact learning) come from the review.

#	REVIEW POINT	IN V2	V3	WHAT V3 ADDS	SECTION
1	Confidence ≠ decision quality; meta decision model	F04 card	Expanded	Full decision pipeline; one meta-model instead of stacked adjustments; four states	Decision engine
2	Stock DNA	F03 card	Expanded	Best setups, best regime, typical holding, failure patterns; sample-size rule	Stock DNA
3	Setup-specific ML	F07 card	Expanded	Per-stock setup probabilities; model catalogue with labels and months	ML models
4	Expected value replaces confidence for ranking	F06 card	Expanded	Worked example; UI shows confidence + Expected R + risk + setup	Decision engine
5	Time-horizon intelligence	F05 card	Expanded	Horizon strip; failing heads hidden	ML models
6	News intelligence is too shallow	F08 card	Expanded	Nine-step pipeline, event list; impact numbers split out as F34	News brain
7	Event shock detector	F09 card	Expanded	Action list for positions, watchlist, stops, candidates	News brain
8	"Why not buy?"	F10 card	Expanded	Why / why not with WATCH → wait for pullback	Decision engine
9	"What changed?"	F11 card	Expanded	Point contributions and the main change	Decision engine
10	Deeper portfolio intelligence	F15–F17 cards	Expanded	Sector + correlation + factor → concentration → health; financials cluster	Portfolio brain
11	Opportunity-cost engine	F14 card	Expanded	Worked example; user decides	Portfolio brain
12	Failure investigator	F31 card	Expanded	Detection rules and loss breakdown; moved to Core	Learning brain
13	Model disagreement	F25 card	Expanded	Agreement count (4 of 6 → WAIT)	Decision engine
14	Regime change as a global event	F26 card	Expanded	Two-day confirmation and six actions	Market brain
15	Personal trading twin	F19 card	Unchanged	Kept in the moat; User Brain in the architecture	12-month moat
16	AI copilot over our own data	F20 card	Expanded	Tool list, cited answers, "would reconsider if"; owner beta Month 6	AI copilot
17	Morning AI briefing	F21 card	Expanded	Full screen mock; top setups by Expected R	AI copilot

#	REVIEW POINT	IN V2	V3	WHAT V3 ADDS	SECTION
18	Monte Carlo	F29 card	Expanded	Outputs and stress list	Research lab
19	What-if simulator	F30 card	Expanded	Sector shock added	Research lab
20	Research lab	F28 card	Expanded	Pipeline; v1 screen moved to Month 6	Research lab
21	Non-ML gaps: multi-user, subscriptions, data, compliance	P1-P4 cards	Expanded	portfolio_id, watchlist_id, broker_connection_id added	Data model
22	Freeze the architecture	Layered diagram only	New in v3	Data Brain + User Brain diagram, adopted with one correction	Architecture
23	Priorities P0 / P1 / P2	Phases 0-5 by gate	Changed	Core / Differentiation / Moat order, run beside the product track	6-month roadmap
+	Advanced portfolio optimisation	Not in v2	New in v3	Added as F33	Portfolio brain
+	Event-impact learning, failure-pattern learning	Inside F08, F31	New in v3	F34 added; failure patterns feed the challenger loop	12-month moat

Where v3 differs from the review

One meta-model, not stacked adjustments

The review adds news, sector and risk as separate steps, such as "Decision impact +7%". Stacked adjustments double-count and break calibration. In v3 every input feeds one meta-model, calibrated on walk-forward folds. The pipeline diagram shows reading order, not maths order.

News impact numbers need history

"Historical similarity 82%, typical 10-day impact +4.1%" needs years of classified announcements, and that history starts the day we record it. Classification ships in Month 5; impact numbers come later as event-impact learning (F34).

The user profile doesn't change a stock's DNA

The review draws User Brain → Stock DNA. DNA is a shared fact about a stock, the same for everyone. In v3 the User Brain feeds personal ranking, portfolio correlation and risk sizing instead.

The priority list skips the gates

The review's P0 is all intelligence. v3 keeps that order for the intelligence track, but on the owner's paper book first. No public calls before SEBI registration and a data licence, and no real-money claims before the edge is proven.

Fundamentals aren't available yet

The review lists Fundamentals in the Data Brain. There is no point-in-time source today; NSE XBRL parsing is a multi-week project in the 12-month plan.

Per-stock percentages need samples

"Breakout 78%" for one stock needs enough past breakouts in that stock. v3 shows them only with 20 or more past setups, pulled toward the sector average, with the sample size on screen.

CURRENT SYSTEM

What SwingTradeML already does

Read from the code in backend/src/swing_trade_ml and docs/ROADMAP_TRACKER.md. These parts stay; v3 builds on them.

AREA	TODAY	WHERE
Features	53 per stock per day: trend, momentum, volatility, volume, price structure, relative strength vs NIFTY and sector, VIX, breadth	ml/features.py
Models	One classifier per cap tier (large, mid, small), LightGBM by default. Barrier models (+8% before −4% in 15 bars) run as shadows	ml/train.py, ml/registry.py
Validation	Purged 5-fold walk-forward, calibration report, realised accuracy on live predictions	ml/calibration.py
Evidence so far	AUC 0.556–0.583. The top 5% of picks earned +0.22R per trade in a simplified test; the full-system replay lost −0.1% to −0.4% per 8-month window after costs	docs/superpowers/research/
Market regime	Six regimes from NIFTY trend, volatility, breadth and VIX, each setting how much capital can be deployed (10% to 85%)	services/deployable.py
Risk	1% risk per trade from stop distance, single-stock and sector caps, cap-tier budget, cash floor, ₹10K to ₹1Cr ladder, 15% drawdown halt	services/risk.py, limits.py
Landmines	No entries near results or corporate actions, or for ASM, GSM or T2T stocks	services/avoid.py
Exits	Stop, target, trailing stop, half out at +5%, 30-day time stop, confidence-based exit	services/exit_policy.py
Data feeds	Kite candles and quotes; NSE delivery %, bulk and block deals, FII/DII, restriction lists, event calendar	services/market_feeds.py
Factors	Trend and low-volatility percentiles; value, quality and size are stubs; not yet wired to the scan	services/factors.py
Execution	Paper and live brokers behind a two-flag latch; every decision logged, rejections included	brokers/, services/execution.py
Research	Backtest replay with the real cost model, over CLI and API; no screen yet	services/backtest.py
Screens	Dashboard, Scan results, Strategies, Models, Model Lab, Positions, Holdings, Capital, Safety, Reports, Real report, Settings	frontend/src/pages
Alerts	Telegram to one chat: trades, errors, 16:00 daily summary	notifications/telegram.py
Users	Password and Google sign-in; trading tables have no per-user owner	core/security.py

Today's pipeline

15:40 Daily candles and NSE files	15:45 · Scan Each strategy scores each stock	Landmines Results, corporate actions, surveillance lists	Risk Sizing, caps, deployable %	Execution Paper or live order; every decision logged	09:15–15:30 Stops and exits every 60 s	16:00 Snapshot and Telegram summary
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What the swing-trading sources say, and what we take from them

Five public sources were read in full. None presents an ML model or backtest results: all are rule-based strategies illustrated with hand-picked charts. Together they give one consistent playbook, and our app already follows most of it. Every rule taken from them must pass our own tests before it goes live.

SOURCE	AUTHOR, DATE	MAIN CONTRIBUTION FOR US
WallStreetMojo	—	Basics: hold days to weeks; buy at support, sell at resistance; Fibonacci, T-line, candlesticks; 2:1 reward to risk
Elearnmarkets	Vivek Bajaj, 23 Jun 2026	Five strategies: trend following, breakout, reversal, pullback, momentum; the 1% risk rule
Investing.com Academy	Dan Blystone, 28 Apr 2026	Confluence of indicators; multi-timeframe agreement; more chart and candle patterns; liquid large caps above the 20 and 50 averages
AlgoTest	AlgoTest, Jun–Jul 2026	Exact rules for five strategies; 1–2% risk; at least 1:2; India execution, liquidity, price above ₹100; 20% short-term tax on delivery gains
BottomStreet	Undated, 2026	A full pullback rulebook: screener, next-day trigger entry, day-6 time stop, breakeven at 1.5R, VIX and market-stretch filters

The consensus playbook compared with SwingTradeML

STEP	WHAT THE SOURCES AGREE ON	SWINGTRADEML TODAY	DECISION
1 · Market first	Index above its 20–50 EMA; healthy breadth; VIX not spiking	Regime table: trend, breadth, VIX percentile	Have + test two new rules below
2 · Sector next	Trade stocks in outperforming sectors; sector index above its 20 EMA	Sector feature, no gate	Add sector brain (F02)
3 · Stock in a trend	Above EMA 20 and 50, 50 rising, higher highs and lows	SMA-based trend features	Add EMA 20/50 and swing structure
4 · A named setup	Pullback (all 5), breakout on volume (4), reversal (3), momentum (2), range (1)	One model scores everything	Add setup playbook below (F07)
5 · A trigger	Wait for confirmation: next day above the trigger candle's high, on volume	Buys at the signal-day close	Test first (F40)
6 · Stop at structure	Below the swing low or inside the broken range	Stop by ATR distance	Test first (F41)
7 · Size by risk	1–2% of capital per trade	1%	Have keep 1%
8 · Reward ≥ 2× risk	Structure targets; part profit; trail	+8% / –4%; half out at +5%; trailing stop	Have

STEP	WHAT THE SOURCES AGREE ON	SWINGTRADEML TODAY	DECISION
9 · Time stop	Exit a stalled trade around day 6	30 calendar days	Test first (F41)
10 · Avoid landmines, keep a journal	No trades near events or in illiquid stocks; log every trade	Blackout 3 days (results), 2 days (actions); journal planned	Test a 7-day blackout

Setup playbook: exact rules for the setup detectors

Starting rules to code and test. Numbers marked * are our proposed starting values, not the sources'. Everything is tuned only through walk-forward testing.

SETUP	STOCK CONDITION	TRIGGER	STOP	TARGET AND EXIT	SOURCES	DECISION
Trend pullback	EMA 20 above EMA 50; 50 rising; higher highs and lows	Bullish candle within 1–2% of the 20 EMA; next day trades above its high	1–2% below the swing low	Part at the prior swing high; trail on the 20 EMA	AlgoTest, Elearnmarkets	Add · Month 2
Quality pullback	5–12% off the swing high; 38.2–61.8% retracement; volume below the 20-day average; delivery above 50%; range narrowing; support tested twice; sector index above its 20 EMA	Candle closes in the upper half; next day +0.25% above its high on ≥ 1.5× volume	Below the swing low (the lower of two)	61.8–100% projection of the prior swing; breakeven at +1.5R; exit day 6 if no move	BottomStreet, Elearnmarkets, Investing.com	Add · Month 2
Breakout on volume	10–15 session base; volatility squeeze	Close above resistance or the 20/55-day high* on 1.5–2× volume; OBV at a new high	Just inside the broken range	Range height projected from the breakout	AlgoTest, Elearnmarkets, WallStreetMojo	Add · Month 2
Momentum	Top 10%* on the trend factor and relative strength; sector leading	New 20-day high* on above-average volume	Below the last swing low	Trail; exit when relative strength rank drops out of the top 30%*	Elearnmarkets, our factors	Add · Month 2
Reversal at support	At tested support after RSI below 30; bullish RSI divergence; long-term trend still up*	Hammer, bullish engulfing or a double bottom confirmed	Below the pattern low	Pattern depth or the next resistance	Elearnmarkets, Investing.com, WallStreetMojo	Test · Month 3
Bollinger + RSI mean reversion	Quality stock above its 200-day average*, sharp drop on short-term news	Touches the lower 20-day band, closes back inside, RSI turns up from below 30	Below the recent low	The 20-day middle band	AlgoTest	Test · Month 3
Range trading	Range-bound large cap; each level tested 2+ times	Rejection candle at support	Just below support	The top of the range	AlgoTest	Later (small ranges rarely reach +8%)

New building blocks the playbook needs

Price structure

- Swing high and low detection
- Support and resistance levels, with how often each was tested
- Fibonacci retracement and projection levels
- Base and range detection (10–15 sessions)

Indicators

- EMA 5, 20, 21, 50 (SMA and EMA 12/26 exist)
- RSI divergence
- Volatility squeeze (Bollinger width at a low)
- Weekly-bar trend for multi-timeframe checks

Patterns

- Candles: hammer, engulfing, shooting star, marubozu
- Charts: double top and bottom, head and shoulders, pennant, cup and handle
- Rule-based detectors with unit tests on hand-built charts

Test list: ideas to prove on our data (Months 1–3)

#	IDEA	SOURCE	HOW WE TEST IT	ADOPT IF
T1	Next-day trigger entry (buy-stop above the signal day's high) instead of buying at the close. Also solves the live blocker on after-close orders	BottomStreet	Episode replay + walk-forward replay	Higher R after costs; fewer false entries
T2	Pullback filters: volume shrinking, delivery above 50%	BottomStreet	Filter scorecard with shadow scoring	Blocked stocks would have lost money
T3	Stop below the swing low instead of ATR distance	All	Replay, same entries	Lower loss per stop with no drop in hit rate
T4	Breakeven stop at +1.5R	BottomStreet	Replay	Better R, fewer full losses
T5	Day-6 time stop for trades that haven't moved	BottomStreet	Replay	Frees capital without cutting winners
T6	No new entries when India VIX is above 20 and rising	BottomStreet	Episode replay, filter scorecard	Protects in weak phases without hurting normal markets
T7	No new entries after NIFTY rises more than 8% in 15 sessions	BottomStreet	Replay	Avoids late-momentum losses
T8	Weekly trend must agree with the daily signal	Investing.com	Walk-forward	Higher top-bucket precision
T9	7-day event blackout instead of 3	BottomStreet	Filter scorecard	Fewer gap losses; few good trades lost
T10	Market breadth on the Nifty 500 (above 60% over their 50 EMA) instead of the watchlist	BottomStreet	Regime replay	Regime calls earlier and steadier

Setup-specific exits (trail on the 20 EMA, structure targets, day-6 time stop) would change the locked trade spec of +8% / −4% / 15 bars. That's your decision, and it's taken only if replay shows the change beats the fixed spec.

Not required, and why

ITEM FROM THE SOURCES	WHY NOT
Day trading and scalping	A different product; our scan and costs are built for multi-day holds

ITEM FROM THE SOURCES	WHY NOT
Futures and options swings	We trade delivery (CNC) only; options also lose value daily. Revisit only after the equity edge is proven
4-hour and 1-hour charts	We scan completed daily bars after the close by design
Forex and crypto	Out of scope
Margin and leverage	Adds risk that the capital ladder is built to avoid
Ease of Movement indicator	Low value; overlaps with volume features. Add only if an ablation shows it helps
Buying at the close ahead of a trigger	The sources warn against it (gap risk); T1 tests the opposite
Claims such as “20–30% faster moves”	No evidence given; we trust only our own tests

Our own additions beyond the sources

Proof before use

Every rule is scored by the setup and filter scorecards and adopted only with out-of-sample evidence. The sources give no evidence at all.

Market memory

Setups are judged per market phase, so the bear-mode shortlist uses only setups that held up in past weak phases.

Expected R ranking

When several setups fire, they're ranked by Expected R after costs, not by how good the chart looks.

Shadow scoring

Every filtered stock is followed, so we learn what each rule costs or saves.

Tax view

P&L after an estimated 20% short-term capital gains tax on delivery trades (F45), confirmed with an accountant.

Same rules for every plan

Setups are computed once; plans only change how much a user sees.

RESEARCH 13. Competitor notes: Tickertape

COMPETITOR · SCREEN RECORDING OF TICKETAPE.IN, 29 SEP 2026

What Tickertape shows, and what we take from it

A 28-second screen recording with no audio, browsing Tickertape's home page, portfolio tools, screener and plans. Tickertape is a research and screening platform (65 lakh+ downloads, 4.2 rating shown), not a signal service, so it's a reference for product design and pricing, not for trading logic.

WHAT THEY SHOW	WHAT IT IS	DECISION FOR TRADEMIND	WHERE
Live ticker strip	Scrolling prices of popular stocks and indices	Add for watchlist and holdings; needs the licensed data feed for users	Home · P4
Market news feed with tags	"Sansera leads losers", tagged by stock, with time	Covered by the news brain, with event type and materiality added	F08
Corporate events (dividend ex-date, results)	Upcoming events for stocks	Have the data (event calendar); show it as a user-facing calendar	F08 · A3
"Latest deals by Radhakishan Damani"	Big named investors' bulk and block deals	Add : we already store bulk and block deals daily and use none of it	F42
Curated screens ("Cash Rich Smallcaps")	Ready-made stock screens	Add our own ready screens: quality pullbacks, volume breakouts, leading-sector momentum, bear-mode safe list	F44
Diversification score vs community (57)	User's score beside the average of all users	Add to portfolio health, using anonymous averages only	F43
Best and worst bets; asset distribution	Portfolio breakdown	Covered by portfolio health and the journal	F15 · F18
smallcase baskets with CAGR	Ready model portfolios with past returns	Later : model portfolios are research services under SEBI rules, and showing CAGR needs verified performance	Moat
Trending Themes (Space Economy, AI Leaders, Semiconductors, Commodities)	Theme baskets of US stocks with price, move and bookmark	Add for Indian themes (Defence, Railways and PSU, EV, Capital goods, Renewable energy, Manufacturing), ranked by momentum like sectors. The US themes themselves are not required	F46
Bookmark on each stock	Add to watchlist in one tap	Have the watchlist; add a one-tap bookmark on every stock card	P1
Pro upgrade page: ₹399 a month, ₹899 for 3 months, ₹2,399.2 a year; coupon UPGRADE20 for 20% off	Billing periods, best-value badge, coupon with savings line	Add the page pattern, with GST shown, honest pricing, and periods limited by SEBI's advance-fee rule; the prices are a benchmark data point	F48
Sign-up pop-up: install QR and app picture on the left; +91 phone and "Get OTP" on the right; terms line	Phone-OTP sign-in with an app-install prompt	Add : same layout; password or Google now, phone OTP at go-live; consent to RA and AI-use disclosures	F49

WHAT THEY SHOW	WHAT IT IS	DECISION FOR TRADEMIND	WHERE
Ads, and “go ad-free with Pro”	Monetisation of the free plan	Not required : third-party financial ads beside our calls create conflicts; our Free plan earns through upgrades	—
Digital gold, US stocks, loans against mutual funds	Other products	Not required : outside Indian equity swing trading	—
Mutual fund screens and portfolio	MF research	Not required now : our MF module is frozen until trading phases land	—
Screener used by 700K+ investors; download and rating badges	Social proof in marketing	Later : allowed only within the SEBI ad code (no “best” or “No 1”)	Compliance

What TradeMind has that Tickertape doesn't show: daily BUY / WATCH / WAIT / AVOID decisions with evidence for and against, Expected R, bear-market memory, score report cards, and position management for what you already own.

Read the market before the stock

Every morning the system builds one market state: regime, index trends, sector rotation, global cues and liquidity. Every model, screen and alert reads that same state.

MARKET · 08:30

Illustrative layout

Regime

● Risk-on · deploy up to 75%

INDEX TREND

NIFTY 50

↑ Strong

BANKNIFTY

↑ Strong

MIDCAP

→ Neutral

SMALLCAP

↓ Weak

INDIA VIX

↓ Falling

SECTOR ROTATION

Capital goods

↑ Leading

Auto

↑ Improving

Pharma

→ Flat

PSU banks

↓ Weakening

OVERNIGHT

US futures · crude · USD-INR

+0.4% · -1.1% · flat

FII / DII previous day

net buy / net buy

INPUT	STATUS	SOURCE
Regime and deployable %	Built	deployable.py
NIFTY, VIX, breadth	Built	market_context.py
BANKNIFTY, MIDCAP, SMALLCAP trends	New	Kite index candles
Sector rotation ranking	Extend	Sector indices, delivery %
FII / DII in the rule table	Extend	Stored daily, unused
Global cues	New	New feed
Liquidity	New	Traded value, spreads

Regime stays a rule table: five years of history holds only a handful of real regime changes, too few to learn from.

Regime change is a system-wide event

A new regime counts only after two days in a row, so one noisy day doesn't flip it. Once confirmed, six things happen and each is logged:

- 1

Re-score every stock
- 2

Re-rank opportunities
- 3

Review open positions
- 4

Recalculate deployable capital
- 5

Review stops
- 6

Notify users (Plus and Pro)

A regime change can alter selection and deployable capital only. Stop, target, risk per trade, sector cap, time stop and the drawdown halt stay locked (roadmap locked decision 3).

Sector brain

Measures

- Sector momentum
- Relative strength vs NIFTY
- Breadth inside the sector
- Money flow from delivery %
- Volatility

Labels

- Leading, improving, flat, weakening, lagging
- Leadership: the top 3 sectors
- Deterioration: 5-day fall in rank

Hierarchy

Market → Sector → Industry → Stock. For example: NIFTY → Industrials → Capital goods → POLYCAB. Industry level needs a classification source; `sector_map.py` maps to NSE sector indices today.

A bear market should be a known situation, not a surprise

The app remembers every past weak phase: how deep and long it was, what our system did, and what it lost. When a similar phase starts, it shows the closest past episode, switches to a tested defensive playbook, and tells each user what it means for their holdings.

Defences already built

- Regime table cuts deployable capital to 30% in a weak market and 10% when stressed (deployable.py)
- Confidence bar raised by 0.10 in a NIFTY downtrend (bear_market_confidence_boost)
- Falling-knife, liquidity and volatility filters (ml_swing.py)
- 15% drawdown halt on new entries, never force-selling
- Sector cap, cash floor, 30-day time stop

What's missing

- **No measurement by market phase.** The replay (May 2023 – Aug 2026) reports averages per 8-month window, never “during the correction we lost X”.
- **Defences not replayed.** The replay skipped the regime deployable %, sector cap and cash floor, so we don’t know how much they protect.
- **No memory.** Nothing stores past episodes or compares today with them.
- **No bear-market experience for users.** Fewer picks appear with no explanation.

1 · Episode library: remember every weak phase

Episodes are found by rule from stored index data, not picked by hand. **Correction:** NIFTY 10% or more below its 52-week high. **Bear market:** 20% or more. **Shock:** 5% or more down in 5 trading days. **Weak phase:** regime weak or stressed for 10+ trading days. Mid- and small-cap indices are checked the same way, because they often fall harder.

STORED FOR EACH EPISODE	EXAMPLES
Shape	Start, bottom and recovery dates; depth for NIFTY, midcap and smallcap; days down; days to recover
Conditions	VIX peak, breadth low, FII / DII flow, leading and lagging sectors, triggering events
What our system did	Trades taken, win rate, R per trade, drawdown; entries blocked by each defence (regime, confidence bar, drawdown halt, filters)
What worked and what failed	Setups that still made money; failure labels of the losses; sectors that held up
What-if	The same period with each defence switched off, and with alternative playbooks

FIRST EPISODES TO LOAD	DATA	STATUS
Late 2024 – early 2025 correction	Inside stored candles and the replay window	Month 1
Late 2021 – mid 2022 correction	Inside the 5 years of stored candles	Month 1
Early 2020 COVID crash	Needs back-fill from the NSE bhavcopy archive (a file per date)	Month 2

FIRST EPISODES TO LOAD	DATA	STATUS
2018 mid- and small-cap fall	Needs back-fill from the NSE archive	Month 2
Every weak phase from now on	Recorded live, point-in-time	Ongoing

Exact dates and depths are computed from stored data by the rules above; this document doesn't guess them.

2 · What the user sees on a bear-market day

BEAR MODE · DAY 12

Illustrative values

Regime

● Risk-off · deploy up to 10%

NIFTY from 52-week high

-11%

Smallcap from high

-19%

Stocks above 50-day average

22%

INDIA VIX

High

FII flow, 10 days

Net selling

CLOSEST PAST EPISODE

Episode B (a 5-month correction)

similar on breadth, FII selling

Then: further fall after day 12

-6% more

Then: days to bottom / to recover

48 / 110

WHAT OUR SYSTEM DID THEN

Trades · win rate · R per trade

14 · 36% · -0.2R

Entries blocked by the regime rule

61

Largest drawdown

-2.1%

Today's playbook

Large caps only · pullbacks in leading sectors · cash is a position

Past episodes are a reference, not a forecast. There are only a few, so no probability is claimed.

YOUR HOLDINGS IN BEAR MODE

Illustrative values

High-beta holdings

3 of 7

In lagging sectors

2

Below stop-loss

1 · review today

Portfolio in cash

18%

IF EPISODE B REPEATS ON YOUR PORTFOLIO

Estimated fall

-9% to -14%

Hardest hit

small-cap holdings

What the brief says instead of silence

"Only 2 picks today because the market is in bear mode. In similar past phases our BUY signals won less often, so the system is holding more cash. Your 3 high-beta holdings need a look."

3 · The playbook by phase

Phases describe where the market is, not where it will go. Each phase uses only the levers the roadmap allows to change: selection and deployable capital. Stop, target, risk per trade, sector cap, time stop and the drawdown halt never change.

PHASE	SIGNS	WHAT THE SYSTEM DOES	WHAT THE USER SEES
Early warning	Regime turns weak; breadth under 40%; FI selling for a week	Deploy up to 30%; confidence bar +0.10; small caps paused; shortlist narrowed	Regime-change alert; “why fewer picks”; holdings with high beta flagged
Decline	Stressed regime; NIFTY 10%+ off its high; VIX in its top 20%	Deploy up to 10%; large caps only; only setups that held up in past episodes	Bear mode banner; closest past episode; daily holdings review
Panic	VIX spike; breadth near its lows; shocks on high volume	No new entries unless a tested playbook allows them; shock detector on every holding	Calm, factual brief; no hype; what happened in past panics
Base and recovery	Breadth rising; regime back to normal for 2 days; leading sectors emerging	Step deployable back up by the rule table; leaders first	“Recovery signs” card: sectors turning first, compared with past recoveries

4 · If it repeats: questions the memory must answer

How much did we lose, and why?

Per episode: system result vs NIFTY, the loss labels, and which setups failed. From the episode replay, not averages.

Which defence saved the most?

Replay each episode with one defence switched off at a time: regime limit, confidence bar, drawdown halt, filters.

What still worked?

Setups, sectors and cap tiers that made money in each episode. They become the bear-mode shortlist.

Would a different playbook do better?

Test alternatives on some episodes, confirm on episodes they weren't tuned on, and check they don't hurt normal markets.

What happens to my portfolio?

Replay a past episode's actual daily moves on the user's current holdings (extends the what-if simulator).

How long did recovery take?

Days to bottom and to recover per episode, shown as history, never as a promise.

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
B1	Detect episodes by rule from stored index data	Every correction of 10% or more in stored history is in the library, with dates and depths
B2	Episode replay of the real system, including the regime deployable %, sector cap and cash floor	Replay output per episode: trades, win rate, R, drawdown, entries blocked by each defence
B3	Defence ablation per episode	A table of result change when each defence is switched off
B4	Closest-episode match for today	Shows the nearest episode, what matches and what differs; no probability claimed
B5	Bear mode from the regime rule with 2-day confirmation	Mode changes are logged; the brief explains them
B6	Playbook uses selection and deployable levers only	Test proves stop, target, risk per trade, sector cap, time stop and halt are unchanged in bear mode

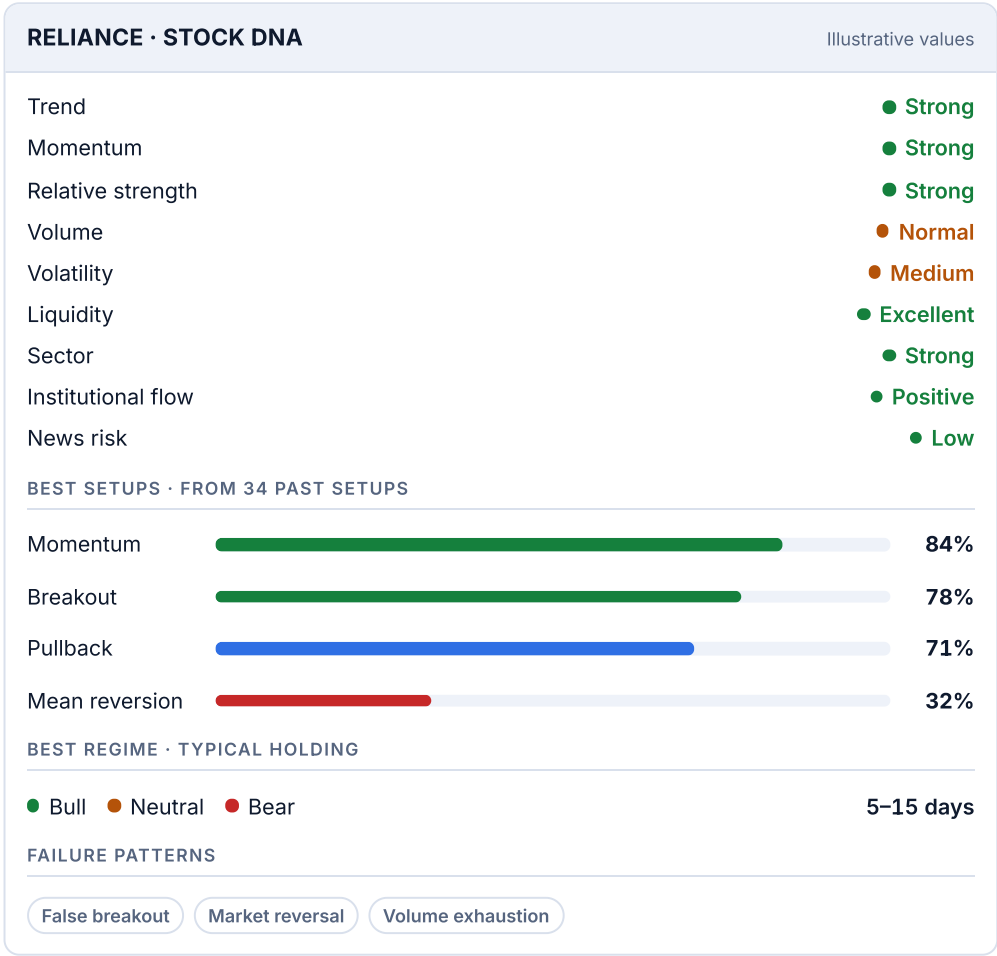
#	REQUIREMENT	ACCEPTED WHEN
B7	A playbook change needs out-of-sample proof	Better on episodes it wasn't tuned on, and no worse in normal markets
B8	Episode repeat on the user's portfolio	Uses real historical daily returns of the held stocks in that episode
B9	No bottom or recovery predictions	Copy review: no “the market will recover in X days” anywhere in the app or copilot

Plans: Free sees the bear-mode banner and one line on the closest episode. Plus adds the episode card and holdings flags. Pro adds the full memory, defence ablations and the episode repeat on their portfolio.

STOCK DNA · F03

A lasting, readable profile for every stock

The 53 daily features become a profile a person can read, recomputed every evening. Traits come from today's data. The learned parts come from our own history with that stock and appear only once there's enough of it.



TRAIT	COMPUTED FROM
Trend	SMA ratios, ADX, trend factor percentile
Momentum	RSI, MACD, 10- and 20-day rate of change
Relative strength	20-day return vs NIFTY and vs its sector
Volume	Volume vs 20-day average, OBV slope
Volatility	ATR %, 20-day volatility percentile
Liquidity	Average traded value, spread
Sector	Sector rank from the sector brain
Institutional flow	Bulk and block deals in 20 days, delivery % trend
News risk	Upcoming events, recent material announcements

Each trait is a cross-sectional percentile against the day's universe: Strong ≥ 70, Normal 30–70, Weak < 30.

Learned sections and their rules

Best setups

Win rate of each setup in this stock, from the trading journal. Shown only with 20+ past setups, pulled toward the sector average for small samples, with n on screen.

Best regime, typical holding

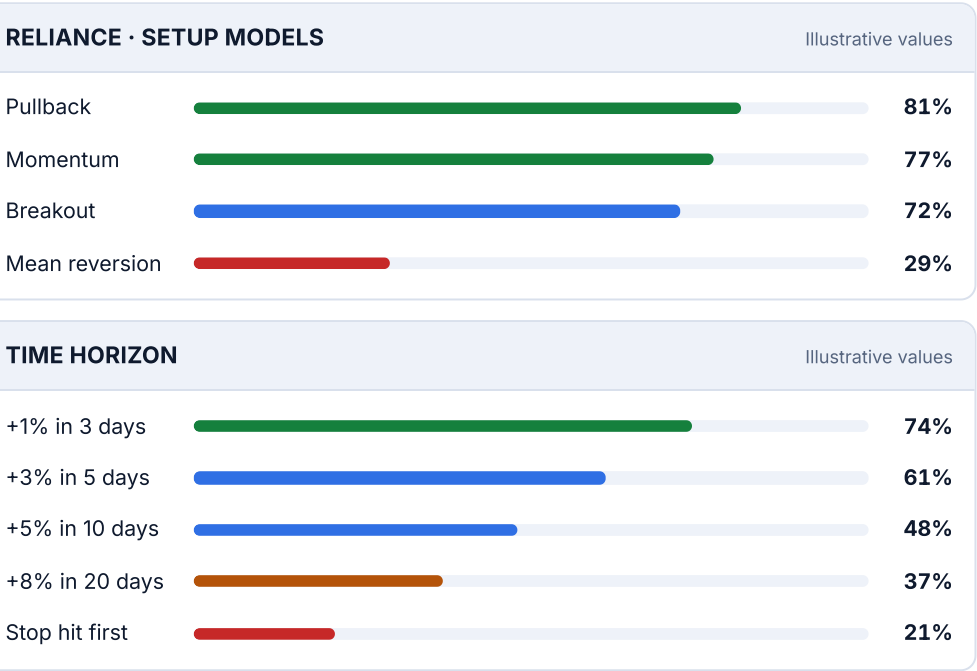
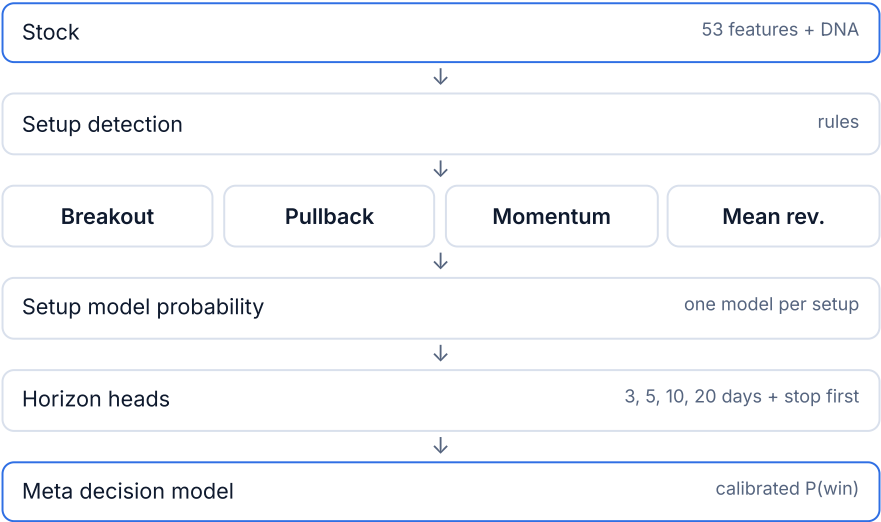
Outcomes of past signals split by regime; median days from entry to target or stop.

Failure patterns

The most common loss labels from the failure investigator for this stock. Appears after the investigator runs (Month 5).

From one model to a family that explains why

Today one classifier scores every stock. v3 first detects which setup a stock is in, scores it with that setup's model, adds time-horizon odds, and calibrates the result. A stock that matches no setup gets no signal.



Model catalogue

MODEL	QUESTION IT ANSWERS	LABEL	STATUS	WHEN
Cap-tier classifiers (3)	Will this stock move up?	Endpoint label; barrier versions in shadow	Live / shadow	Built
Breakout model	Does this breakout reach +8% before −4%?	Barrier, breakout candidates only	New	Month 2
Pullback model	Does this pullback resume the trend?	Barrier, pullback candidates only	New	Month 2
Momentum model	Does momentum continue?	Barrier, momentum candidates only	New	Month 2

MODEL	QUESTION IT ANSWERS	LABEL	STATUS	WHEN
Mean-reversion model	Does an oversold stock bounce?	Barrier, oversold candidates only	New	Month 3, kept only if it beats the baseline
Horizon heads (4)	+1%/3d, +3%/5d, +5%/10d, +8%/20d	One barrier label per horizon	Extend	Month 4
Stop-first head	Is -4% hit before the target?	Inverse barrier	Extend	Month 4
Meta decision model	Combined, calibrated P(win)	Logistic over sub-scores, isotonic calibration	New	Month 3
Exit model	Hold, reduce, trail or exit?	Post-exit outcome label	Extend	Month 5
Event classifier	What kind of announcement is this?	Language model, fixed event list	New	Month 5
Event-impact model	How does this event type usually move the stock?	Returns after past classified events	Future	After 12 months of events

Rules every model must pass

Beat the baseline

It must beat today's classifier on purged walk-forward folds and on the paper stream. Sector, breadth, VIX and delivery % each failed this once.

Calibrated or hidden

Calibration gap ≤ 10 points in every populated bucket. A head that fails is hidden from users, not shown with a warning.

Human promotion

It runs as a shadow first, then goes live only by a recorded human decision with one-call rollback.

Read announcements, not just dates

Today the system knows when results and corporate actions are due and blocks entries near them. It doesn't read what companies announce. The news brain reads every NSE filing and turns it into a structured event the decision engine can use.



EVENT · XYZ LTD		Illustrative values
Event	Large order received	
Sentiment	Positive	
Materiality	High · order ≈ 9% of revenue	
Horizon	5–20 days	
Filing	NSE link	
AFTER 12 MONTHS OF RECORDED EVENTS		
Similar past events	n = 41	
Typical 10-day move	shown with range	
Impact numbers stay hidden until there's enough recorded history to measure them.		

EVENT SHOCK · 10:07		Illustrative values
Stock	XYZ	
Price	+4.3%	
Volume	5.2× average	
Linked event	New order announcement · 10:08	
ACTIONS		
Existing position	REVIEW	
Watchlist	RE-RANK	
Stop	RE-CALCULATE	
New candidates	RE-CHECK	

Event list

EVENT	WHERE IT COMES FROM	HOW IT'S USED AT LAUNCH
Results, earnings guidance	NSE filings, XBRL	Blackout before results (built); tone of guidance
Order win, order cancellation	NSE filings	Materiality vs revenue; sentiment

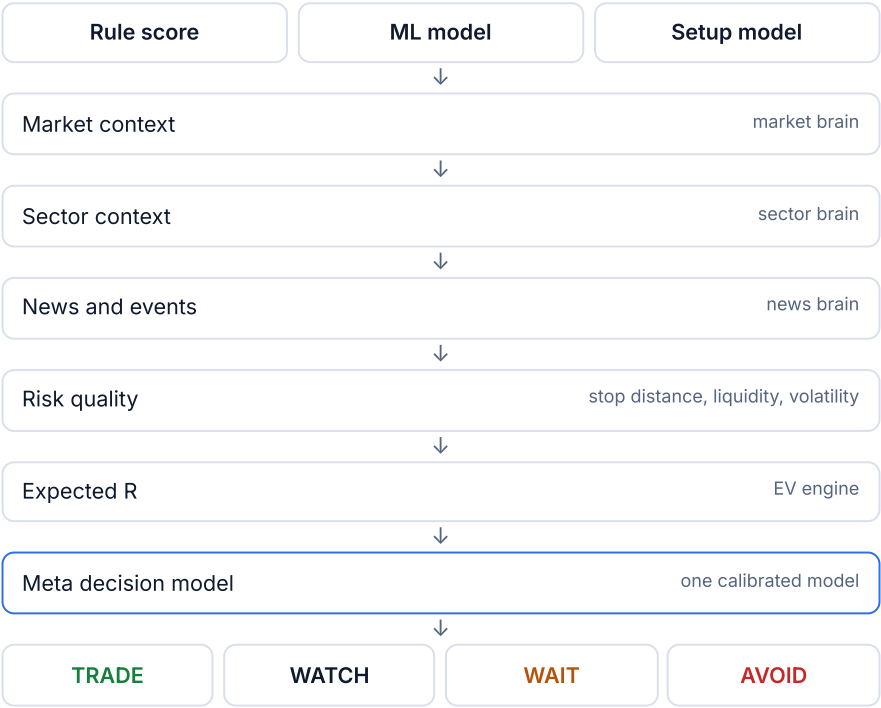
EVENT	WHERE IT COMES FROM	HOW IT'S USED AT LAUNCH
Management change	NSE filings	Flag; caution on key exits
Regulatory action, litigation	NSE filings	Negative flag; can force AVOID
M&A	NSE filings	Flag; high volatility expected
Dividend, buyback, split, bonus	Corporate actions	Blackout around the date (built)
Promoter pledge, insider trade	SAST / PIT disclosures	Pledge increase is negative; insider buying positive
Credit-rating change	NSE filings	Direction-based flag
Government and sector policy	Press releases	Sector-level flag
Macro events (RBI, budget, CPI)	Calendar	Market-level caution day

A language model classifies each filing into this list and scores materiality and sentiment. It never decides a trade, and every event links to its filing. Accuracy must reach 85% on 200 hand-labelled filings before release.

DECISION ENGINE · F04, F06, F10, F11, F22, F25

Confidence is not decision quality

One number can't say whether a trade is worth taking. The decision engine gathers every input and ends in one of four states, each with its evidence. Users see confidence, Expected R, risk and setup side by side, never one magic score.



STATE	WHEN
TRADE	All gates pass and Expected R is above the threshold. Shown as BUY.
WATCH	A good stock, but the entry is poor (for example 3%+ above the entry zone) or it needs confirmation.
WAIT	Uncertain: models disagree, the regime is changing, data quality is low, or an event is due.
AVOID	A landmine, weak or negative Expected R, or poor risk quality.

The flow shows reading order. In the maths, every input enters one meta-model together, so nothing is counted twice and the output stays calibrated.

Rank by Expected R, not confidence

EV (in R) = P(win) × average win - P(loss) × average loss - costs

STOCK	P(WIN)	AVG WIN	AVG LOSS	EXPECTED R	RANK
C	61%	+2.8R	-1R	+1.32R	1
A	68%	+2.1R	-1R	+1.11R	2

STOCK	P(WIN)	AVG WIN	AVG LOSS	EXPECTED R	RANK
B	78%	+1.2R	-1R	+0.72R	3

Worked example, before costs. B has the highest confidence but the lowest edge. (The review's table showed +1.10R and +1.11R; the formula gives +1.11R and +1.32R.)

RELIANCE · WHY AND WHY NOT

Illustrative values

WHY BUY

✓ Strong trend

✓ Sector leadership

✓ Relative strength vs NIFTY

✓ Volume confirmation

✓ Setup model 72% · Expected R +0.9

WHY NOT BUY

! Market volatility rising

! Entry 3.2% above the ideal zone

! Results in 4 days

Decision

WATCH · wait for pullback

RELIANCE · WHAT CHANGED

Illustrative values

Yesterday

84

Today

76 (-8)

WHY

Sector strength

-4

Volume

-2

Relative strength

-1

News

-1

Main change

Sector leadership weakened

MODEL AGREEMENT

Illustrative values

Rule engine

BUY

Breakout model

BUY

Momentum model

BUY

News

BUY

Risk

WAIT

Regime

BEAR

Agreement 4 of 6

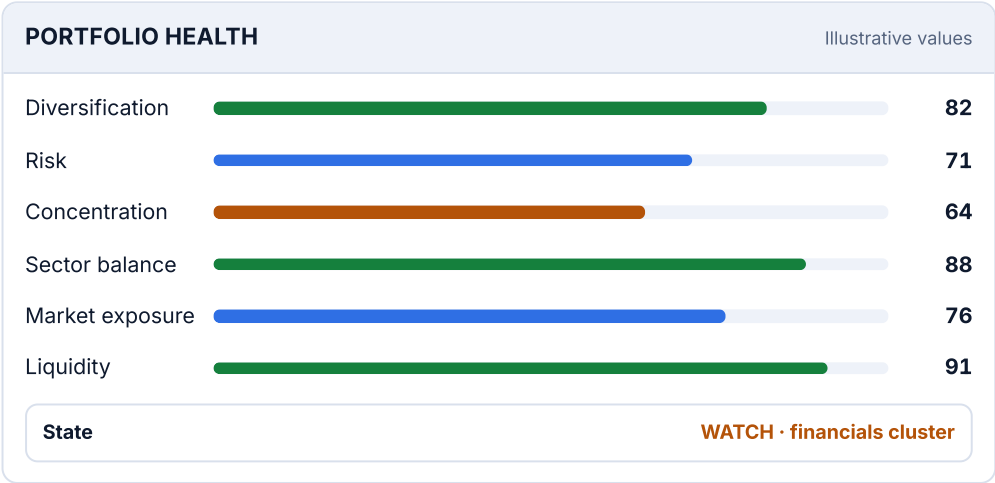
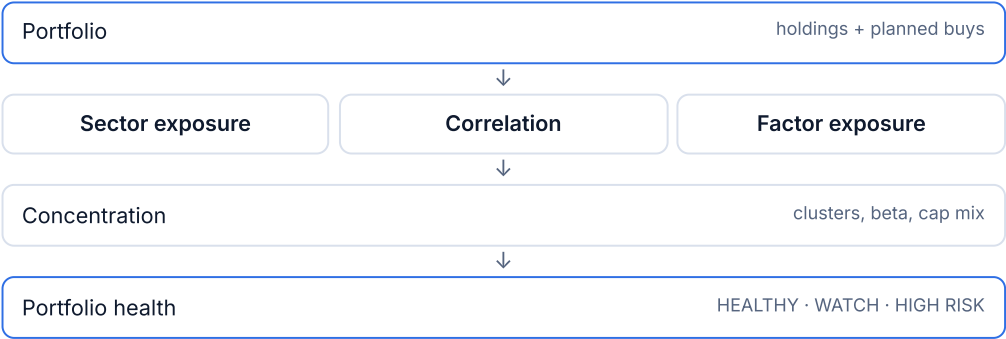
WAIT

The system says "I don't know" instead of forcing a BUY.

PORTFOLIO BRAIN · F14, F15, F16, F17, F33

Six stocks can be one bet

Sector caps exist today. Correlation doesn't, and factor exposure is only partly built. The portfolio brain looks at sector, correlation and factor exposure together, finds concentration, and gives one health state.



The hidden cluster

A bank, a finance company and an NBFC carry three different sector labels. Their 120-day return correlation can still be above 0.7, which makes them one financials bet. The correlation engine groups them and warns before a fourth joins.

Six auto names (Tata Motors, M&M, Eicher, Maruti, Bharat Forge, Ashok Leyland) are the obvious case; the sector cap already catches that one.

What it measures

- Stock correlation: 60 and 120 days
- Sector and industry correlation
- Factor exposure: momentum, volatility, size, liquidity now; value, quality, growth after fundamentals
- Beta concentration
- Market-cap concentration

OPPORTUNITY COST		Illustrative values
ABC (held) · remaining edge		+0.35R
XYZ (new setup) · expected edge		+1.10R
Difference		+0.75R
Sector concentration after swap		Acceptable
State		REVIEW ABC vs XYZ
Shown only when the gap is 1R or more in the live rule; the user decides. Nothing is sold automatically.		

Advanced portfolio optimisation F33 · moat

Suggests weights across today's TRADE candidates and current holdings to get the most Expected R within the locked risk limits, the correlation limit and the user's cap split. The output is a proposal the user accepts or edits. It builds on the diversification calculator.

Needs the EV engine, correlation engine and 6+ months of calibrated outcomes, so it sits in the 12-month plan.

An interface to the intelligence, not a replacement for it

The morning brief gives users a reason to open the app. The copilot answers “why” questions using our own structured data through read-only tools, never general web knowledge.

TRADEMIND · 08:45

Illustrative values

Market

Risk-on

NIFTY · BANKNIFTY · MIDCAP · VIX

↑ · ↑ · → · ↓

SECTOR LEADERS · WEAKNESS

Defence, Banking, Energy

leading

IT, Pharma

weak

TODAY

Major announcements

3

Your holdings needing attention

2

Stocks avoided (risk or event)

5

TOP SETUPS BY EXPECTED R

1. XYZ · pullback

+1.18R

2. ABC · breakout

+0.94R

3. PQR · momentum

+0.81R

Strategy today

Pullback + momentum · avoid illiquid breakouts

COPILOT

Illustrative answer

You: Why is HAL no longer a good candidate?

Copilot: HAL's score moved from 82 to 68 since Monday. Main reasons:

1. Defence sector strength declined

2. Relative strength vs NIFTY fell

3. Volume confirmation disappeared

4. Market volatility increased

5. Results are due in 5 days

Current state

WATCH

It would be reconsidered if:

sector strength recovers, volume confirms, and the score rises above 78.

Sources: decision record, what-changed record, event calendar.

Copilot tools

TOOL	RETURNS	EXAMPLE QUESTION
market_today	Regime, index trends, sectors, events	What changed overnight?
stock_dna	Traits and learned sections	How does RVNL usually behave?

TOOL	RETURNS	EXAMPLE QUESTION
decision	State, evidence for and against, thresholds	Why is this WAIT and not BUY?
changes	Score changes with contributions	Why is RVNL appearing today?
portfolio_health, correlation	Health scores, clusters	Which of my holdings needs attention?
whatif	Portfolio impact of a shock or a new buy	If I add this stock, how does my risk change?
journal, failures	User's trades, loss labels	Show my worst-performing setup this month.
strategy_stats	Stats by regime	How does breakout do in weak markets?
similar_trades	Past trades with matching context	Find trades similar to this setup.

Guardrails: tools are read-only, every answer cites its data, and there are no personal order instructions (“buy 200 shares”) until a legal opinion clears it. It's Pro only, with a daily question cap, and every answer is logged for audit. The owner uses it first, in Month 6.

Ask why the strategy fails, not only whether it made money

Every decision goes into the journal with its full context. Losing trades get a failure label. The model observatory compares models, and challengers replace the champion only with human approval.



LAST 100 LOSSES		Illustrative values
False breakout		28%
Market reversal		21%
Late entry		17%
News shock		12%
Exit timing		11%
Model error		7%
Execution		4%

LABEL	DETECTION RULE (FIRST VERSION)
False breakout	Breakout entry closes back below the breakout level within 3 bars, then stops out
Market reversal	Regime turned weaker, or NIFTY fell 3%+, while held
Late entry	Entry more than 1 ATR above the setup trigger
News shock	Material event while held, with a gap through the stop
Exit timing	Price reached the original target within 5 days of exit
Execution	Fill slippage more than twice the cost model
Model error	High confidence with no rule or context explanation for the loss

Model observatory scorecard

MEASURE	WHY
AUC and top-bucket precision	Ranking skill
Calibration gap by bucket	Whether the stated probability is true
Realised R per trade after costs	Profitability
Maximum drawdown	Pain
Results by regime	Where it works and where it doesn't
Drift: feature shift, outcome by bucket	When it stops working

Never allow a newly trained model to replace production because it scores better on one backtest. The registry already supports versions and one-call rollback.

EXPERIENCE ENGINE · F37, F38, F39

Every score, factor and filter earns its place from its own track record

After a few months the app should know from its own evidence which scores, factors and filters work, and in which market. That evidence comes from two places: history replayed over five years of stored data (available now) and the app's own daily testing (grows every day).

A score can't make returns mandatory. Markets can't be guaranteed, and SEBI prohibits promising returns. What we make mandatory is **accountability**: every score band publishes what it actually delivered, and a band that stops delivering is flagged automatically. “When the app says 80+, here is what 80+ has actually done.”

1 · Score report card · F37

Today calibration.py buckets confidence against outcomes, and evaluate_pending_signals scores every signal against its target, stop and horizon. The report card turns that into a promise the user can check.

SCORE REPORT CARD · LAST 90 DAYS

Illustrative values

SCORE	SIGNALS	TARGET FIRST	AVG R	STATUS
90+	18	—	—	Building record
80–89	142	58%	+0.41R	Delivering
70–79	236	49%	+0.12R	Delivering
60–69	311	38%	–0.08R	Under review

Market during these 90 days

mostly Normal, 21 days Weak

After costs. A band shows numbers only after 30 resolved signals. Tabs split the same card by regime.

Rules

- Every band shows its sample size, the share that hit the target first, average R after costs, and its calibration gap, overall and per regime.
- Below 30 resolved signals a band shows “building record”, not a number.
- Every day, the last 60 resolved signals of each band are checked. If the real hit rate falls more than 10 points below the stated one, the band is marked “under review” and the app stops showing it as a BUY.
- Refitting the score is a challenger change and needs human approval. Flagging is automatic; changing is not.
- Users see the card in the app; any public claim uses PaRRVA-verified numbers only.

2 · Filter and factor scorecard · F38

Each filter blocks stocks, and each factor selects them. The scorecard follows the stocks a filter blocked and asks what would have happened. A filter that blocks losers is protecting us. A filter that blocks winners is costing us money.

FILTER SCORECARD · WEAK MARKETS			Illustrative values
FILTER	BLOCKED	BLOCKED WOULD HAVE MADE	VERDICT
Regime capital limit	61	-0.31R	Protecting
Falling-knife filter	44	-0.42R	Protecting
Bear confidence bar +0.10	38	-0.05R	Neutral
Results blackout	12	—	Building record
Liquidity filter	57	+0.22R	Costing us
<i>"Blocked would have made" is the average R those stocks reached against the same target, stop and horizon, after costs.</i>			

FACTOR SCORECARD · BY REGIME				Illustrative values
FACTOR	NORMAL	WEAK	STRESSED	
Trend	+0.35R	+0.02R	-0.20R	
Low volatility	+0.08R	+0.21R	+0.30R	
Sector strength	+0.28R	+0.15R	—	
Relative strength	+0.31R	+0.05R	—	
Spread in R between the top and bottom fifth of stocks ranked by each factor. “—” = building record.				

TRACKED ITEM	TODAY	WHAT'S NEEDED
Risk and execution rejections (slots, caps, drawdown halt)	Stored with a reason and already scored against target, stop and horizon	Group outcomes by reason and regime
Filters inside the strategy (liquidity, volatility, falling knife)	Turn a candidate into HOLD with no stop or target, so they can't be scored	Shadow scoring: record a shadow signal with the stop and target it would have had
Landmines (results, corporate actions, ASM/GSM/T2T)	Refused with a plain reason	Shadow scoring, same as above
Regime limit and bear confidence bar	Applied, never measured	Shadow scoring plus the episode replay (F35)
Factors (trend, low volatility; later value, quality)	Scored in factors.py, not wired	Daily factor spread by regime

Verdicts need 50 or more blocked signals per filter per regime. A verdict produces a proposal (keep, tune, drop) for the owner. It never removes a locked safety rule by itself; the drawdown halt and sector cap stay, and their scorecards only inform review.

3 · The lessons loop: better every bear market · F39

When a weak phase ends (recovery confirmed by the regime rule), the app writes a lessons report and proposes playbook changes. A change is adopted only if it helps on past episodes it wasn't built from.

1 · Episode ends Regime back to normal for 2 days	2 · Lessons report What we lost, why, which filters saved us, what still worked	3 · Proposals Shortlist, factor weights, deployable steps, setups to pause	4 · Out-of-sample test Replay on the other stored episodes	5 · Normal-market check No damage outside bear phases	6 · Approval Playbook version goes up by one
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What “successful” means in a bear market

Lose much less than the market, lose less than in the last similar episode, and keep capital ready for the recovery.
Profit in a falling market is welcome but not the target.

Measured every episode

- Our fall vs NIFTY's fall (target: less than half)
- Loss vs the previous comparable episode
- Cash available at the bottom
- Days to regain the previous high

Applied next time

In bear mode the shortlist favours the factors, setups and sectors whose scorecards were positive in past weak phases. The closest-episode card shows which lesson is being applied.

4 · How experience builds up

WHEN	MARKET EXPERIENCE (HISTORY REPLAY)	OUR EXPERIENCE (LIVE DAILY TESTING)
Month 1	Five years replayed: scorecards and episode results for 2021–22 and 2024–25	Shadow scoring switched on; every decision and every block recorded
Month 3	2018 and 2020 back-filled and replayed	First live score bands and busy filters pass their minimum sample in the current regime
Month 6	Playbook v1 tested out-of-sample across episodes	Filter verdicts and factor spreads in the regimes seen so far
Month 12	Every new episode added and replayed	Live records across more regimes; live evidence weighs more than replay where both exist

The months are estimates. Records fill at the speed of signals and regimes, so a quiet month adds less. The card always shows the sample size, and nothing is stated until its minimum is reached.

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
E1	Score report card by band and regime	Every band shows n, target-first rate, avg R after costs, calibration gap
E2	Minimum sample before any number	Bands under 30 and filters under 50 show “building record”
E3	Daily band health check	A band more than 10 points under its stated rate is flagged and stops showing as BUY the same day
E4	Shadow scoring of every filtered candidate	Each filter and landmine block leaves a scoreable record with stop, target and horizon
E5	Filter scorecard	For each filter and regime: blocked count, what blocked stocks did, verdict
E6	Factor scorecard	Daily top-minus-bottom spread per factor and regime
E7	Episode lessons report	Written automatically when a weak phase ends
E8	Changes need out-of-sample proof and approval	No score, filter or playbook change goes live without both
E9	Locked safety rules are never removed by a scorecard	Test: drawdown halt, sector cap, stop and risk per trade can't be changed by any verdict
E10	No return promises	Copy review: the report card says “has delivered”, never “will deliver”

Test ideas before real money

The backtest engine and API already exist with the real cost model. The lab adds a screen, then walk-forward, Monte Carlo and what-if analysis as queued jobs.

Choose
Strategy, universe,
date range, regime

Rules
Entry, exit, costs,
slippage, sizing

Backtest
Existing engine

Walk-forward
Out-of-sample folds

Monte Carlo
Bad-luck scenarios

Paper
Shadow run

Champion / challenger
Human approval

MONTE CARLO · ₹10,00,000

Layout only

Outcome range (5th–95th percentile)

from your trades

Worst simulated drawdown

—

Chance of a drawdown above 20%

—

Longest losing streak

—

Capital survival

—

STRESS ADDED TO RESAMPLED TRADES

Losing streaks

Slippage

Gaps

Volatility spikes

Market crash

Higher correlation

Values come from the user's own trade history; none are shown here.

WHAT IF?

Layout only

SCENARIOS

NIFTY –5%

VIX × 2

Small cap –10%

Banking –8%

RESULTS

Estimated portfolio impact

₹ and %

Impact per stock

by beta and correlation

Risk increase

health change

Positions to review

list

An estimate from betas and correlations, not a forecast.

Version 1 (Month 6) puts a screen on the existing backtest. Version 2 (Months 7–12) adds walk-forward, Monte Carlo, what-if and job quotas per Pro user.

PLANS AND MONETIZATION

Free discovers, Plus decides, Pro understands

Free

Discover · ₹0

- 5 large-cap picks a day, HIGH / MEDIUM / LOW
- Regime label and short brief
- Top reason a stock isn't a BUY
- End-of-day stop alerts
- Paper trading, strategy catalogue

Plus

Decide · about half of Pro

- Full list with sub-scores and Expected R
- Why and why not, what changed (watchlist)
- Index and sector trends, regime-change alerts
- Real-time alerts, holding action states
- Health score, 1 model and 1 strategy

Pro

TradeMind · 50% below market

- Stock DNA, setup models, horizon odds, EV ranking
- News brain, shock alerts, exit intelligence
- Portfolio brain: correlation, factors, opportunity cost
- Copilot, personal brief, Research Lab, Model Lab
- Later: trading twin, what-if, Monte Carlo

Pricing rule from the plan doc: about 50% below comparable products, Plus at about half of Pro, and Pro free for 6 months at launch if the SEBI check allows it. Exact prices wait for a market check.

Feature by plan

✓ with a note = included with that limit · — = not included. Platform and internal items are left out.

FEATURE	FREE · DISCOVER	PLUS · DECIDE	PRO · TRADEMIND
Market			
Regime and deployable capital A1	✓ Regime label	✓ Regime + deployable %	✓ Full rule view
Market intelligence layer F01	✓ Risk-on / Neutral / Risk-off	✓ + NIFTY, BANKNIFTY, MIDCAP, SMALLCAP trends	✓ + global cues, liquidity
Sector brain F02	—	✓ Sector ↑ → ↓	✓ + industry, leaders, deterioration
Regime change detector F26	—	✓ Alert	✓ + re-rank and re-risk
Market memory (episode library) F35	✓ Closest episode, one line	✓ Episode card: what happened, what we did	✓ Full memory + defence ablations
Bear mode and playbook F36	✓ Bear-mode banner	✓ + holdings flags, why fewer picks	✓ + phase playbook, episode repeat on your portfolio
Big-investor deals F42	✓ Top deals of the day	✓ All deals + your watchlist	✓ + alerts when a deal hits your holdings
Theme rotation F46	✓ Top 3 trending themes	✓ Theme ranks + their stocks	✓ + theme momentum alerts and theme filter on setups
Portfolio			
Risk engine and capital ladder A2	—	✓ Fixed 1% sizing	✓ Custom limits
Exit policy A6	✓ SL HIT at end of day	✓ Real time	✓ Real time + trailing

FEATURE	FREE · DISCOVER	PLUS · DECIDE	PRO · TRADEMIND
Position intelligence F12	—	✓ Action state	✓ + thesis tracking
Exit intelligence F13	✓ SL HIT (end of day)	✓ HOLD / REDUCE / TRAIL / EXIT	✓ + exit reasons and score
Opportunity cost engine F14	—	—	✓ Held vs new edge
Portfolio health F15	—	✓ Health score	✓ Six scores + clusters
Correlation engine F16	—	—	✓ Clusters, beta, cap concentration
Factor exposure F17	—	—	✓ Price factors now; fundamental later
What-if simulator F30	—	—	✓ NIFTY, VIX, cap, sector shocks
Advanced portfolio optimisation F33	—	—	✓ Suggested weights to accept or edit
Structure stops, targets and time stop F41	—	✓ Stop and target levels	✓ + breakeven and time-stop status
Community diversification benchmark F43	—	✓ Your score vs community	✓ + by capital band
Tax view F45	✓ Estimated tax on paper trades	✓ On your trades	✓ + yearly tax report export
Decision			
Landmine filter A3	✓ Applied to every pick	✓ Applied, with reasons	✓ Applied, with reasons
Meta confidence system F04	✓ HIGH / MEDIUM / LOW	✓ 5 sub-scores + Expected R	✓ + calibration and model detail
Expected value engine F06	—	✓ Expected R per pick	✓ Ranking by Expected R
“Why not buy?” engine F10	✓ Top reason	✓ Why and why not	✓ + what would change it
“What changed?” engine F11	—	✓ Watchlist	✓ All + portfolio
Explainability F22	✓ Top 3 reasons	✓ For and against	✓ + feature contributions
Data quality F24	✓ Badge	✓ Badge	✓ Per-source detail
Uncertainty and model agreement F25	✓ “Low confidence: wait”	✓ + reasons	✓ + agreement count
Trigger entry F40	—	✓ Trigger price on each pick	✓ + trigger status alerts
Learning			
Calibration, walk-forward, Model Lab A5	—	✓ Summary	✓ Full
Trading journal F18	✓ Paper trades	✓ Trades + outcomes	✓ Full context + failure labels
Model observatory F23	—	✓ Summary	✓ Full lab

FEATURE	FREE · DISCOVER	PLUS · DECIDE	PRO · TRADEMIND
Strategy lab F27	✓ Read-only catalogue	✓ Choose 1	✓ All + stats
Research lab F28	—	—	✓ Backtest v1; walk-forward and Monte Carlo later
Monte Carlo stress testing F29	—	—	✓ Drawdown range, survival
Failure investigator F31	—	—	✓ Loss labels and totals
Score report card F37	✓ What each label has delivered	✓ Report card by score band	✓ + by regime and calibration detail
Filter and factor scorecard F38	—	—	✓ Which filters and factors are working, by regime
Lessons loop F39	—	✓ Lesson being applied	✓ + full lessons report per episode
Stock			
Stock DNA F03	—	✓ 4-trait summary	✓ Full profile, setups, regime, holding, failures
Time-horizon probabilities F05	—	—	✓ 3, 5, 10, 20-day odds + stop first
Setup-specific models F07	—	✓ Choose setups	✓ Per-setup probabilities
News classification F08	✓ Event badges	✓ Event list	✓ + materiality, sentiment, horizon
Event shock detector F09	—	—	✓ Real-time shock alerts
Event-impact learning F34	—	—	✓ Similar events and typical move
Ready-made screens F44	✓ 2 screens	✓ All ready screens	✓ + custom screens
Stock page F47	✓ Header, decision label, daily chart, top reason	✓ + Expected R, setup, trigger, alerts, intraday chart, full evidence	✓ + DNA, setups and odds, peers, overlays, full record
User			
Personal trading twin F19	—	—	✓ Personal ranking
AI copilot F20	—	—	✓ Cited answers over your data
Morning brief F21	✓ Short	✓ Full	✓ + your portfolio

Success measures

METRIC	TELLS US	FIRST TARGET
Realised R per pick after costs	Whether the edge exists	Above 0 over 3 months and above the SMA benchmark
Calibration gap	Whether “67%” means 67%	≤ 10 points per populated bucket
EV ranking vs confidence ranking	Whether Expected R ranks better	Higher realised R in the top decile

METRIC	TELLS US	FIRST TARGET
Brief open rate	Daily habit	≥ 40% of active users on trading days
Free → Plus → Pro conversion	Plan value	Set after the launch offer ends
30-day retention by plan	Product fit	Tracked from day one

USER CONTROLS

The user sets the inputs; the system does the work

Market intelligence is computed once for everyone. The user layer then filters, ranks, sizes and alerts using what each person tells us. We never place an order for a user.

1 · Shared

Market, sectors, news, DNA

2 · Shared

Decisions with evidence

3 · Your plan

What you see, and when

4 · Your choices

Models, strategies, watchlist

5 · Your money

Capital, risk, holdings

6 · Your alerts

Channels, quiet hours

Risk profile at sign-up

All plans

User does

Answers 5 questions: capital, largest acceptable loss, holding period, caps, experience.

System does

Sets a starting preset, risk per trade and alert channels.

User sees

Editable defaults.

Guardrail

Never above 3% risk per trade or past the 15% drawdown halt.

Diversification calculator

Plus presets · Pro any value

User does

Enters capital, reserve, amount to deploy, cap split, number of stocks, weighting.

System does

Fills each cap bucket from today's TRADE list; sizes at the smaller of allocation and risk limit; checks correlation.

User sees

A day plan with quantities and a health score.

Guardrail

Planning tool from the user's own inputs, not personal advice.

Holdings

Plus · Pro

User does

Links a broker read-only or uploads a holdings file.

System does

Stores the thesis, runs position and exit intelligence.

User sees

HOLD, MONITOR, REDUCE or EXIT with reasons.

Guardrail

We never sell for the user.

Model and strategy choice

Plus 1 each · Pro any

User does

Picks models and strategies.

System does

Filters the shared scan; nothing re-runs per user.

User sees

Picks from that mix with evidence.

Guardrail

Untested combinations are labelled; risk rules stay locked.

Watchlist and alerts

All plans

User does

Adds stocks, channels, quiet hours.

System does

Watches what changed, regime and shocks.

User sees

Alerts on chosen channels.

Guardrail

Free alerts at end of day.

Journal and twin

Pro

User does

Marks signals taken or skipped; logs own trades.

System does

Learns preferences after 50+ decisions.

User sees

Personal ranking and patterns.

Guardrail

Opt-in consent; delete any time.

Copilot

Pro

User does	Asks questions.
System does	Calls read-only tools and cites data.
User sees	Answers with sources and “would reconsider if”.
Guardrail	No personal order instructions; daily cap.

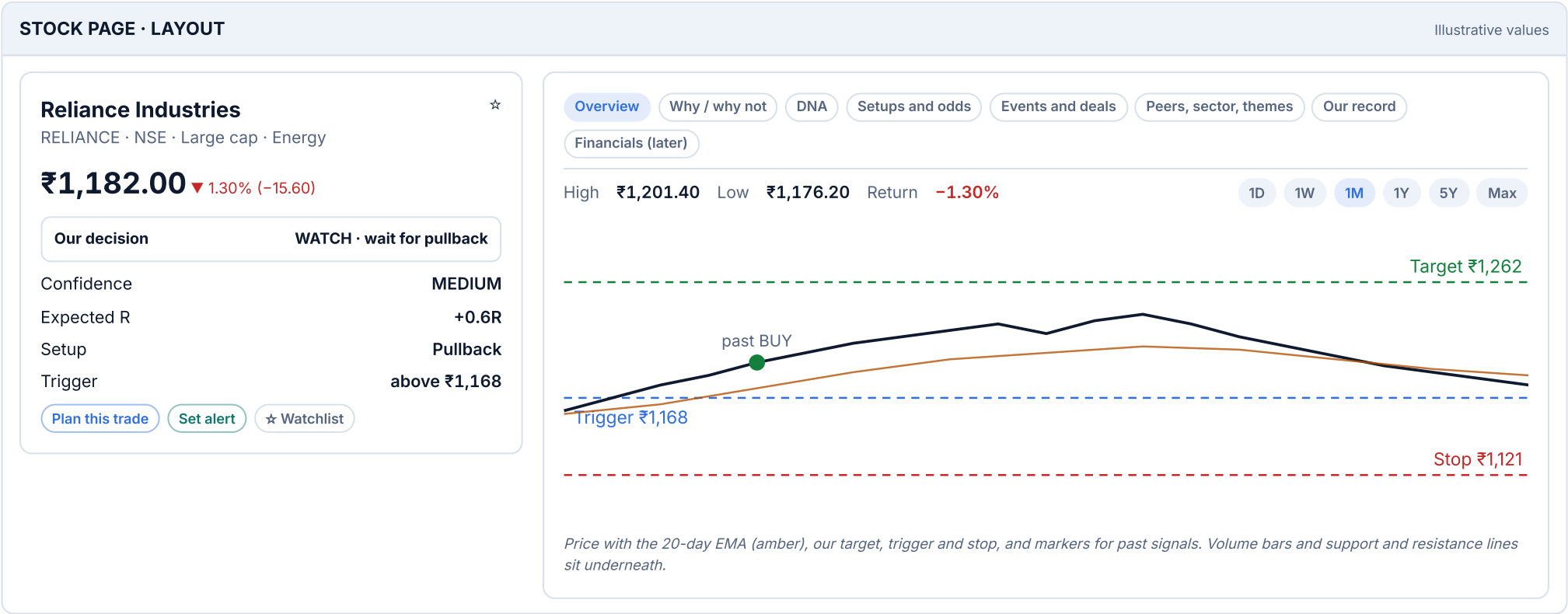
Research lab and what-if

Pro

User does	Sets strategy, universe, costs, or a shock.
System does	Queues backtests, Monte Carlo, scenario maths.
User sees	Results with out-of-sample checks.
Guardrail	Quotas; labelled as estimates.

Everything about one stock, with our decision first

When a user opens a stock, they get one page: a header card with price and our decision, a chart with our levels on it, and tabs for every piece of intelligence. The layout follows the familiar pattern from the Tickertape page you shared (header card, tabs, range buttons), built our own way. Today's StockDetailModal shows the card details and a daily closing-price chart with a lookback picker; this page replaces it.



Page parts

PART	WHAT IT SHOWS	FREE	PLUS	PRO
Header card	Name, symbol, exchange, cap tier, sector, themes; price and day change; watchlist bookmark	✓	✓	✓
Our decision	TRADE / WATCH / WAIT / AVOID, confidence label, Expected R, setup, trigger price	Label (large caps in Free picks)	+ Expected R, setup, trigger	+ horizons, agreement
Actions	Watchlist, set a price or signal alert, plan this trade (calculator with this stock), share	Watchlist	+ alerts, plan	+ order-list export
Chart	1D, 1W, 1M, 1Y, 5Y, Max; high, low, return; previous-close line; hover price and time	Daily, end of day	+ 1D intraday	+ overlays
Chart overlays	Target, trigger and stop lines; EMA 20/50; support and resistance; past signal markers; volume	Stop and target	+ EMAs, past signals	+ support and resistance, setups
Tab · Why / why not	Evidence for and against, what changed since yesterday	Top reason	Full	+ what would change it
Tab · DNA	Traits, best setups, best regime, typical holding, failure patterns	—	4 traits	Full
Tab · Setups and odds	Setup-model probabilities, time horizons, stop-first odds	—	—	✓
Tab · Events and deals	Classified announcements with filing links; results and ex-dates; big-investor deals	Badges	List	+ impact
Tab · Peers, sector, themes	Sector and theme rank, peers compared on trend, relative strength, volatility	Sector name	Ranks	+ peer table
Tab · Our record	Past signals on this stock and what happened; score history	PaRRVA-verified summary only	Past signals	+ score history
Tab · Financials	Results and ratios, after NSE XBRL parsing	12-month plan		
Footer	RA disclaimer, AI-use note, data time stamp	✓	✓	✓

Where we differ from the example, and why

IN THE EXAMPLE	IN TRADEMIND	WHY
Big Buy and Sell buttons	“Plan this trade” and alerts; broker order only later, with the user confirming each order at their broker	We don't place orders for users; a Buy button next to our call reads like an instruction
“Forecast” tab	“Setups and odds”, showing calibrated probabilities with their track record	No price targets presented as predictions; SEBI bans assured-return language
Currency toggle (\$ / ₹)	Not needed	NSE stocks only, in rupees
Plain price chart	Chart with our levels and past signals	The chart should explain our decision, not just show price
Share button	Share a link; the shared page keeps the disclaimer	Advertising code applies to anything shared

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
SP1	One page per stock, opened from any list, card or search	Same page from Home, screens, watchlist, holdings and search
SP2	Decision first	State, confidence, Expected R, setup and trigger visible without scrolling on phone and desktop
SP3	Chart ranges 1D to Max, with high, low, return and previous close	1D needs intraday data: owner uses Kite; users need the licensed feed, so Free shows daily bars only
SP4	Our levels on the chart	Target, trigger and stop lines match the decision record exactly
SP5	Tabs gated by plan	A Free token receives no Plus or Pro fields for any tab
SP6	One-tap watchlist and alerts	Bookmark syncs to the watchlist at once
SP7	Loads fast	First view under 2 seconds on a phone; tabs load when opened
SP8	Disclaimer and data time stamp on every view and shared link	Compliance check passes

Build order when implementation starts: header card and decision, chart with levels, then the Why / why not tab, then the other tabs as their features land. Each tab shows “coming soon” rather than empty data until its feature passes its gate.

One pop-up for signing up and signing in

Based on the Tickertape pop-up you shared: tapping “Sign up”, or any locked feature, opens one pop-up. The left half invites users to install the app; the right half signs them in. Ours starts with username and password or Google, and switches to phone OTP at go-live, as decided.

SIGN-UP POP-UP · LAYOUT

Our design · illustrative



Scan to install TradeMind

Android, iOS, or add to home screen

Know what to buy, what to avoid, and why

Daily market brief, evidence for every pick, and alerts on your phone.

TradeMind

×

NOW · PRIVATE BETA

Username or email

Password

Sign in

Continue with Google

AT GO-LIVE

+91 ▾ Phone number

Get OTP

6 digits · valid 5 min

By continuing you agree to the Terms, Privacy policy and the Research Analyst disclosures, including how AI is used.

What happens after sign-up

1 · Sign up

Password or Google now; phone OTP at go-live

2 · Consent

Terms, privacy, RA and AI-use disclosures accepted and recorded

3 · Risk profile

Five questions set the starting preset

4 · Free plan

Home with the brief and 5 picks

5 · Upgrade

When a locked feature is tapped, or from the plans page

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
SU1	One pop-up opens from “Sign up”, “Sign in”, or any locked feature	After sign-in the user returns to where they were
SU2	Install panel with a QR code for the app, beside the form	QR opens the right store or the web app; hidden on phones
SU3	Now: username and password, or Google	Works with today's sign-in; invite-only during the beta (invites from the admin panel)
SU4	Go-live: phone with +91 and a one-time code; email code as a fallback	Code valid 5 minutes; resend after 30 seconds; lockout after 5 wrong tries; SMS templates registered on DLT
SU5	Consent on sign-up	Links to Terms, Privacy, RA disclosures and AI-use disclosure; acceptance stored with time and version
SU6	One account per person	Phone, email and Google identities link to one user; duplicates refused
SU7	Straight into onboarding	New users see the risk-profile questions before Home
SU8	Abuse protection	Rate limits per phone and IP on sign-up and OTP requests

A clear upgrade page: plan, billing period, coupon, final price

Based on the Tickertape upgrade page you shared: billing periods with a per-month price, a “best value” option, a coupon box and a clear “you saved” line. Ours adds Plus and Pro side by side, tax shown up front, and the rules a SEBI-registered service must follow.

Market benchmark from the screenshot

TICKERTAPE PRO (SEEN 29 SEP 2026)	BILLED	PER MONTH	NOTE
1 month	₹399	₹399	Charged monthly
3 months	₹899	≈ ₹300	Billed every 3 months
12 months	₹2,399.2	₹200	“Best value”; shown against ₹250
Coupon UPGRADE20	−20%	≈ ₹199.93	“You saved ₹599.8”

Tickertape is a research and screening platform, not a daily-signal service, so it's one data point for the market check, not the whole benchmark. Applying our 50%-below rule to it alone would put Pro near ₹200 a month (monthly) and ₹100–125 a month (yearly). That may not cover the data licence and language-model costs, so check it against the cost table before fixing prices. Offers change often; re-check before launch.

UPGRADE · CHOOSE YOUR PLAN

Illustrative prices

Plus · Decide

Pro · TradeMind ✓

○ 1 month

Charged monthly

₹P1

○ 3 months

Billed every 3 months

₹P3 / month

● 12 months

Best value

₹P12 / month

Billed yearly · only if SEBI allows a 1-year advance

Coupon

LAUNCH20 · applied

GST (18%)

shown separately

You pay today

₹X incl. GST · you save ₹Y

Cancel any time; unused months refunded pro-rata, with no cancellation fee. Launch offer: Pro free for 6 months, then this plan (if allowed).

Page rules

- Plus and Pro on one page, with a link to the full plan comparison.
- Each billing period shows the amount billed and the per-month equivalent.
- A struck-out price only when it was really charged before; no fake “was” prices.
- GST shown clearly, and the final amount shown before payment.
- Coupon box with a clear savings line and the coupon's end date.
- Cancellation and pro-rata refund terms shown before payment.
- No countdown timers or fake urgency; SEBI's advertising code bars misleading pressure.

Check before offering longer periods: SEBI's Dec 2024 amendment allowed Research Analysts to charge at most one quarter in advance, and SEBI proposed in Feb 2025 to allow up to one year. Until the consultant confirms the current limit, offer 1-month and 3-month plans only.

Coupons in the admin panel

SETTING	OPTIONS
Code and discount	Percentage or flat ₹ amount
Applies to	Plus, Pro or both; which billing periods; new or existing users
Limits	Start and end date; total uses; one use per user
Tracking	Redemptions, revenue given up, conversions
Controls	Pause or end a coupon; every change logged in the admin audit log

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
C01	Plan and billing-period choice	Every option shows billed amount and per-month equivalent; totals match the payment gateway
C02	Billing periods follow SEBI's advance-fee limit	Periods longer than the confirmed limit can't be sold
C03	Coupons	Validated on the server; expired, used-up or wrong-plan codes are refused with a clear message
C04	Tax shown	GST line and final amount shown before payment; invoice has GST details
C05	Honest pricing	No struck-out price that was never charged; no countdown timers
C06	Refund terms before payment	Cancel any time; pro-rata refund; no cancellation fee
C07	Launch offer	6 months of Pro, then the chosen plan, only if the consultant confirms it; no card needed during the free period
C08	Admin coupon management	Create, pause and end coupons; redemptions tracked; actions logged

Everyone has their own account. You run the whole product from one panel.

You are the owner and admin. From the admin panel you manage users and subscriptions, including your own; decide what each plan includes; approve model and playbook changes; and control the system. Every admin action is logged.

Roles

ROLE	WHO	CAN	CAN'T
Owner	You (one account, can't be removed)	Everything: users, subscriptions, plan catalogue, approvals, system and safety controls, compliance exports	See users' broker tokens; place orders for users
Admin	Future staff	Users, subscriptions, invites, refunds	Change the plan catalogue, approve models, touch system or trading controls
Support	Future staff	Look up a user's account and plan to answer questions	Change billing; see holdings unless the user grants access
User	Everyone else	Their own account, plan and data	Anything in the admin panel

Today's `users.is_superuser` flag becomes the `role` column; your account is migrated as Owner. The role is checked on the server for every `/admin/*` call, not only hidden in the screen.

Changing your own subscription

You can change your own plan at any time, in two ways. Your owner rights never depend on your plan.

ADMIN · MY ACCOUNT

Illustrative

Role

Owner

Current plan

Pro · complimentary (owner)

Broker link

Zerodha · read-only · linked today 06:12

VIEW AS (TEMPORARY)

Free

Plus

Pro ✓

Shows every screen exactly as that plan sees it, with a "Viewing as Free" banner. Your subscription doesn't change.

CHANGE MY PLAN (REAL)

New plan

Plus

Takes effect

Now

Reason

Testing the upgrade flow

Result

Subscription row updated · logged

View as: a quick look

Switches what you see to Free, Plus or Pro for your session only. Nothing is saved, no alerts change, and the banner shows you're viewing as another plan. Use it to check each plan's screens.

Change my plan: a real change

Updates your subscription row, with the source "owner". Alerts, limits, emails and upgrade prompts all behave as they would for a real user. Use it to test upgrade and downgrade journeys end to end.

Your own trading is separate

Your product plan has no effect on the trading core. The bot's paper and live trading, your Kite session and your safety switches work the same on any plan.

Screens

Users

- List and search: name, email, plan, status, sign-up date, last active
- Invite, suspend, reactivate
- Delete on request, with data-protection deletion

Subscriptions

- Each user's plan, source (paid, trial, launch offer, complimentary, owner), start, end, next bill
- Grant, change, extend a trial, apply the launch offer, cancel
- Refund pro-rata through the payment gateway, with no breakage fee

Plan catalogue

- What each plan includes, as switches per feature
- Limits: picks a day, watchlist size, copilot questions, research jobs
- Prices and trial length; versioned, live without a redeploy

Invites and beta list

- Replaces 600GLE_ALLOWED_EMAILS in .env, which can't be edited in production
- Invite by email; choose the starting plan
- Turn open sign-up on at launch

Approvals inbox

- Model promotions (champion / challenger)
- Playbook versions from the lessons loop
- Filter and score changes from the scorecards
- Approve or reject with a note; each decision logged

System and safety

- Today's Safety page: kill switch, exits switch, halt and resume
- Feed health, job status, data quality
- Owner only

Compliance

- Audit log search and export (5 years)
- Disclaimer and MITC text, versioned
- PaRRVA export, advert register, consent records, deletion requests

Coupons

- Create percentage or flat-₹ codes for Plus, Pro, periods and user groups
- Dates, total uses, one use per user
- Redemptions and revenue given up; pause or end any time

Revenue

- Active subscriptions by plan; trials ending this week
- Monthly revenue, churn, failed payments
- Complimentary and owner plans shown apart, never counted as revenue

ADMIN · SUBSCRIPTIONS

Illustrative

USER	PLAN	SOURCE	RENEWS
you@owner	Pro	Owner	—
tester1@...	Plus	Complimentary	31 Dec
user42@...	Pro	Launch offer	12 Sep
user57@...	Free	—	—

Action on user42

Change plan · Extend offer · Cancel · Refund

Rules for plan changes

- Upgrades take effect now. Downgrades take effect at the end of the paid period unless the admin chooses “now”, which triggers a pro-rata refund.
- A paid feature is never taken away in the middle of a period by a catalogue change.
- A price change applies to new subscriptions; existing users are told in advance.
- Every change needs a reason, and the user is notified.
- Complimentary plans have an end date, so they never become free forever by accident.

Sign-in: now and at go-live

PHASE	USERS SIGN IN WITH	OWNER AND ADMINS SIGN IN WITH	STATUS
Now, through the private beta	Username and password, or Google	Same; the owner account is protected by role checks	Built (password and Google sign-in exist; the Google list moves from .env to invites)
At go-live	OTP sign-in (mobile or email one-time code), alongside Google	Okta single sign-on for the owner and staff, with its own two-factor	Planned

OTP at go-live

One-time codes by SMS or email, with rate limits and lockout after repeated failures. SMS in India needs sender and template registration on the telecom DLT platform, which takes lead time, so start it in the month before launch.

Okta at go-live

Single sign-on for the admin panel, so staff access is added and removed in one place with enforced two-factor. The design keeps sign-in providers pluggable (the `user_auth_providers` table already exists), so Okta is an addition, not a rewrite.

Until then

Strong passwords, Google sign-in, invites managed in the admin panel, and the owner role guarded on the server. No outside users see live calls before go-live anyway.

Requirements and acceptance

#	REQUIREMENT	ACCEPTED WHEN
AD1	Roles: owner, admin, support, user	Every /admin/* call checks the role on the server; a user token gets 403
AD2	Owner account protected	Owner can't be deleted, suspended or demoted. Password or Google sign-in until go-live; Okta with two-factor from go-live
AD11	Sign-in providers: password and Google now; OTP and Okta at go-live	Adding OTP or Okta needs no change to accounts, roles or plans; OTP has rate limits and logout
AD3	Change any user's plan, including your own	Subscription row updated, user notified, action logged with a reason
AD4	View as Free, Plus or Pro	Screens match that plan exactly; banner shown; nothing saved
AD5	Plan catalogue editable without a deploy	Changing a limit takes effect on the next request; catalogue versions kept
AD6	Invites in the database	GOOGLE_ALLOWED_EMAILS no longer needed; invite list managed in the panel
AD7	Approvals inbox	No model, playbook, score or filter change goes live without an approval row
AD8	Admin audit log	Who, what, before, after, reason, time; kept 5 years; can't be edited
AD9	Privacy limits	No admin can read broker tokens; holdings visible to support only with the user's consent
AD10	Refunds follow the rules	Pro-rata, no breakage fee, through the payment gateway, logged

SUBSCRIPTIONS AND COSTS

What we must pay for

Only the Kite price comes from the project's own documents. Other amounts are marked approximate or need a quote; get quotes in Month 1.

ITEM	NEEDED FOR	WHEN	COST	TYPE
Market data				
Kite Connect data plan	Owner's candles, quotes, orders	Now	₹500 a month	Monthly
Licensed data vendor (NSE-authorised)	Prices and live signals shown to users	Month 5	Quote needed	Monthly or yearly
Global cues feed	Market brain	Month 1	Free to low; paid if free feeds fail	Monthly
News API (optional)	News beyond NSE filings	Later	Quote needed	Monthly
Regulation and legal				
NISM Series XV exam	RA qualification	Month 1	About ₹1,500 per attempt (approx.)	One-off; renew every 3 years
RA deposit in a scheduled bank	SEBI RA registration	Month 2	₹1 lakh to ₹10 lakh, by client count	Held, not spent
SEBI registration and BSE enlistment fees	RA registration	Month 2	Confirm with a consultant	One-off + periodic
Compliance consultant or securities lawyer	Registration; opinions on sizing, copilot, free offer	Month 1	Quote needed	One-off + retainer
PaRRVA verification	Showing any past performance	Before marketing	Confirm with the agency	Periodic
Data-protection (DPDP) review	Consent and personal data	Month 3	Quote needed	One-off
Technology				
Hosting: managed Postgres, Redis, workers	Multi-user service	Month 4	Estimate after a load test	Monthly
Claude API (language model)	Brief wording, news classification, copilot	Months 2, 5, 6	Usage-based; measure in a prototype and cap per user	Per use
Payment gateway	Plan billing	Month 4	About 2% per transaction (approx.; confirm)	Per payment
Push (Firebase Cloud Messaging)	Alerts	Month 4	Free	—
Email provider	Summaries	Month 4	Small, per email	Monthly

ITEM	NEEDED FOR	WHEN	COST	TYPE
WhatsApp Business messaging	Pro alerts	Month 6	Per conversation (Meta pricing)	Per use
SMS OTP provider and DLT registration	OTP sign-in at go-live	Month before launch	Per SMS; quote needed	Per use
Okta	Single sign-on for owner and staff at go-live	Launch	Per user per month; quote needed	Monthly
Google Play / Apple developer accounts	Store apps	At store launch	About \$25 one-off / \$99 a year (approx.)	One-off / yearly

Pro's language-model features cost money for every question. Keep a daily copilot cap per user and check cost per Pro user monthly against the Pro price.

LIMITS

What we can't do, or can't do yet

Not possible is a hard limit. **Not yet** needs data, time or users first. **Needs legal check** depends on a lawyer's opinion.

REQUEST	STATUS	WHY	WHAT WE DO INSTEAD
Guaranteed or “accurate” predictions, promising profits	Not possible	Markets are uncertain; SEBI bans promises of assured returns	Calibrated probabilities with a verified track record
Serving live prices and calls to many users from the owner's Kite session	Not possible	Personal-account data is for own use; showing it to others needs a licence	Licensed vendor (P4); private beta until then
Public hit rates or returns without verification	Not possible	SEBI requires PaRRVA verification	Verify through PaRRVA first
Trading users' money, or automatic orders in their accounts	Not possible	Portfolio-management and algo rules	Users place orders; basket export; broker connect with per-order confirmation later
Stacking news, sector and risk as % adjustments on the probability	Not possible	Double-counts inputs and breaks calibration	All inputs enter one calibrated meta-model
A model that retrains and replaces itself nightly	Not possible	Silent regressions	Champion / challenger with human approval
Learning the market regime with ML	Not possible	Too few regime changes in five years	Rule table with change detection
Scraping Screener, Tickertape, Chartink or TradingView	Not possible	No public APIs; terms restrict it	Manual cross-checks only
Copilot or twin telling a user “buy 200 shares with your ₹10 lakh”	Needs legal check	May fall under Investment Adviser rules	Explain our research; calculator uses user inputs; legal opinion
6-month free Pro offer	Needs legal check	No specific rule found; must appear in the MITC	Confirm with the consultant
News “similarity 82%, typical impact +4.1%” from day one	Not yet	Event history starts when recording starts	Classification in Month 5; impact (F34) after about 12 months
Value, quality and growth factors	Not yet	No point-in-time fundamentals	Price factors now; XBRL in the 12-month plan
Per-stock setup percentages for every stock	Not yet	Too few past setups per stock	Shown only with 20+ setups, pulled to the sector average
Tick-level shock detection for everyone at launch	Not yet	Needs a websocket and a licensed real-time feed	60-second polling first, Pro only
Personal twin for new users	Not yet	Needs 50+ of the user's own decisions	Generic ranking until then; opt-in

REQUEST	STATUS	WHY	WHAT WE DO INSTEAD
A score that makes returns mandatory ("80+ means you will earn X")	Not possible	Returns can't be guaranteed; SEBI bans assured-return claims	Score report card: what each band has delivered, with sample size, flagged when it falls short
Predicting the bottom or the recovery date of a bear market	Not possible	Nobody can; the app would be making a return promise	Show how past episodes went, as history, with their range
Probabilities like "70% chance this bear lasts 3 more months"	Not possible	Only a handful of past episodes exist	Show the closest episode and how it differs, with no probability
Reliable Sharpe from a 6-month paper run	Not yet	Sample too short	Shown with sample-size warnings (built)

Data Brain and User Brain feed one decision pipeline

Shared facts flow down the centre. Personal context joins only where ranking, portfolio and risk are decided. Every decision ends in the journal, which feeds the learning loop back to the models.



A exists **B** exists, extend **C** new **D** future Dashed purple: learning loop, only through human approval

Engineering rules carried over from the codebase

Risk rules stay locked

Stop, target, risk per trade, sector cap, time stop and the 15% drawdown halt can't be changed by any profile, plan or user setting. Enforced in types (`extra="forbid"`).

Off by default, byte-identical

Every new layer ships behind a flag. With the flag off, the scan output must be byte-identical to today's, proved by a snapshot test.

Point-in-time data

Every feed is stored with its fetch date. News and order-book history start the day we start recording.

Walk-forward or it doesn't ship

A new model must beat the current one on purged walk-forward folds and on the paper stream.

One meta-model

News, sector, regime and risk enter one calibrated model as inputs, never as stacked percentage adjustments.

Failures are visible

A stale feed raises an alert and lowers data quality. Low data quality means no decision.

DATA MODEL

Shared facts once, personal data by owner

Market tables have no user column and are computed once a day. Personal tables hang off users through a portfolio, watchlist or broker connection, and every /me/* route filters by the signed-in user.



Migration from today

- The owner becomes user 1. The bot's paper and live books become system portfolios owned by user 1.
- positions, trades, orders, strategies and signals gain portfolio_id (and, through it, the user).
- Existing rows are back-filled to user 1, so every current report still reads the same numbers.
- broker_connections replaces the single broker_sessions log. Each holds its own encrypted token, marked read-only or trading.
- User choices of model and strategy live in user_strategy_choices. They filter the shared scan, never re-run it.

Tables

TABLE	OWNER	KEY COLUMNS	HOLDS	FEATURE
Shared market intelligence				
market_state_daily	Shared	date	Regime, index trends, global cues, data quality	F01
regime_events	Shared	date, from, to	Confirmed regime transitions	F26
market_episodes	Shared	episode_id, kind, start, bottom, recovery	Depth, duration, VIX, breadth, flows, sectors	F35
episode_replays	Shared	episode_id, system_version, variant	Trades, R, drawdown, entries blocked per defence	F35, F36
playbook_runs	Shared	date, phase	Bear-mode phase and levers applied	F36
score_band_stats	Shared	date, band, regime, window	n, target-first rate, avg R, calibration gap, status	F37
filter_scorecard_daily, factor_scorecard_daily	Shared	date, filter or factor, regime	Blocked count, counterfactual R, factor spread, verdict	F38

TABLE	OWNER	KEY COLUMNS	HOLDS	FEATURE
signals.is_shadow (new column)	Shared	signal_id	Marks shadow records left by filters, so they're scored but never traded	F38
lessons_reports, playbook_versions	Shared	episode_id, version	Lessons, proposals, test results, approval	F39
sector_rank_daily	Shared	date, sector, industry	Scores and ranks	F02
stock_profile_daily	Shared	date, instrument_id	DNA traits, setup affinity with sample size	F03, F11
setup_detections	Shared	date, instrument_id, setup	Detected setups and trigger levels	F07
decision_records	Shared	date, instrument_id, model_version	Sub-scores, P(win), horizons, Expected R, evidence, state, agreement, data quality	F04–F06, F10, F22, F24, F25
event_items, event_impacts	Shared	filing_id, instrument_id	Classified announcements, measured impact	F08, F34
shock_events	Shared	ts, instrument_id	Intraday price and volume shocks	F09
factor_scores_daily, correlation_snapshot_daily	Shared	date, instrument_id	Factor percentiles, correlation matrix	F16, F17
model_evaluations, experiments	Shared	model_id, window	Scorecards, challenger runs	F23, F32
Per user				
users	User	user_id	Identity, plan, risk profile, consent	P1
portfolios	User	portfolio_id, user_id, kind (paper / real / bot)	A book of positions	P1
watchlists, watchlist_items	User	watchlist_id, user_id	Watched stocks	P1
broker_connections	User	broker_connection_id, user_id, broker, mode	Encrypted token, read-only or trading	P1
user_strategy_choices	User	user_id, strategy_id	Chosen models and strategies	P2
trade_journal	User	portfolio_id, decision_id	Context at entry, outcome in R, failure label	F18, F31
position_thesis, portfolio_health_daily	User	portfolio_id	Entry thesis, health scores	F12, F15
research_runs	User	user_id, run_id	Backtest, walk-forward, Monte Carlo jobs	F28, F29
user_profile_model, copilot_sessions, briefings	User	user_id	Twin model, chat history, personal brief	F19–F21
subscriptions, entitlements, payments	User	user_id	Plan, trial, launch offer, refunds	P2
audit_log, consents, notification_prefs	User	user_id, ts	5-year record, data consent, channels	P3, P5
Admin				

TABLE	OWNER	KEY COLUMNS	HOLDS	FEATURE
users.role (replaces is_superuser)	User	owner / admin / support / user	Access to the admin panel	P6
subscriptions.source	User	paid / trial / launch_offer / complimentary / owner	Why a user has their plan; comps never counted as revenue	P2, P6
plan_catalog	Admin	version, plan, feature, limit	What each plan includes; versioned	P6
invites	Admin	email, starting plan, status	Beta list, replacing GOOGLE_ALLOWED_EMAILS	P6
coupons, coupon_redemptions	Admin / user	code, kind, value, plans, periods, dates, limits	Discount codes and who used them	F48
invoices	User	user_id, period, amount, GST, coupon	Invoices with tax details	F48
admin_actions	Admin	actor, target, action, before, after, reason, ts	Append-only admin audit log, 5 years	P6
approvals	Admin	kind, subject, decision, note	Model, playbook, score and filter change decisions	P6, F23, F39

APIS

One API, gated by plan

Routes stay under `/api/v1` with the existing JWT sign-in. An entitlement check in `api/deps.py` decides for each plan which fields come back, how delayed they are, and the rate limit. Free and Pro call the same endpoints; Free gets less.

METHOD AND PATH	RETURNS	FREE	PLUS	PRO	FEATURE
Market					
GET <code>/market/today</code>	Regime, index trends, global cues	Regime only	+ trends	Full	F01
GET <code>/market/sectors</code>	Sector ranks and labels	—	Arrows	+ industry	F02
GET <code>/market/regime-events</code>	Regime-change history	—	✓	✓	F26
GET <code>/market/memory</code>	Bear mode, phase, closest episode, what we did then	Banner, one line	Episode card	+ ablations	F35, F36
GET <code>/market/episodes/{id}</code>	One past episode with replay results	—	Summary	Full	F35
POST <code>/me/whatif (episode)</code>	A past episode replayed on the user's holdings	—	—	✓	F36
GET <code>/scorecards/scores</code>	Score report card by band and regime	Per label	By band	+ regime, calibration	F37
GET <code>/scorecards/filters</code>	Filter and factor scorecards	—	—	✓	F38
GET <code>/market/episodes/{id}/lessons</code>	Lessons report for an episode	—	Lesson applied	Full	F39
GET <code>/briefing/today</code>	Morning brief	Short	Full	+ personal	F21
Stocks and decisions					
GET <code>/signals</code>	Today's decisions with sub-scores and Expected R	5 large cap, label	Full list	+ EV sort, horizons	F04–F06
GET <code>/stocks/{symbol}/decision</code>	State, evidence for and against, agreement	Top reason	Full	+ what would change it	F10, F22, F25
GET <code>/stocks/{symbol}/dna</code>	Stock DNA	—	4 traits	Full	F03
GET <code>/stocks/{symbol}/setups</code>	Setup-model probabilities	—	—	✓	F07
GET <code>/stocks/{symbol}/events</code>	Classified events	Badges	List	+ impact	F08
GET <code>/changes/today</code>	What changed, with contributions	—	Watchlist	All	F11
GET <code>/alerts/shocks</code>	Event shocks	—	—	Real time	F09

METHOD AND PATH	RETURNS	FREE	PLUS	PRO	FEATURE
Me					
GET /me/portfolios, /me/watchlists	User's books and lists	✓	✓	✓	P1
POST /me/broker-connections	Link a broker (read-only first)	—	✓	✓	P1
GET /me/positions/intel	Action states, thesis health	—	States	+ thesis	F12, F13
GET /me/portfolio/health	Health scores, clusters, factors	—	Score	Full	F15–F17
GET /me/opportunity-cost	Held vs new edge	—	—	✓	F14
POST /me/whatif	Scenario impact	—	—	✓	F30
GET /me/journal, /me/failures	Journal and loss labels	Paper only	Trades	+ failures	F18, F31
POST /copilot/ask	Cited answer	—	—	Daily cap	F20
POST /research/runs	Queued backtest, walk-forward, Monte Carlo	—	—	Quota	F28, F29
Models and platform					
GET /ml/models/{id}/scorecard	Model observatory	—	Summary	Full	F23
GET /strategies/performance (exists)	Strategy stats, PaRRVA-verified when public	Catalogue	1 strategy	All	F27
/billing/*	Plans, payments, refunds	✓	✓	✓	P2
GET /system/data-quality	Feed freshness	Badge	Badge	Detail	F24
Admin (role-checked, not plan-gated)					
GET /admin/users, POST /admin/invites	Users and invites	Owner, admin; support read-only			P6
PUT /admin/users/{id}/subscription	Change any plan, including the owner's; reason required	Owner, admin			P6
POST /me/view-as	Session-only plan preview	Owner, admin			P6
GET /admin/plans, PUT /admin/plans	Plan catalogue, versioned	Owner			P6
GET /admin/approvals, POST /admin/approvals/{id}	Approvals inbox	Owner			P6
GET /admin/audit, /admin/revenue	Audit search and revenue	Owner; admin read-only			P6

Testing the surface

TEST	RULE
Isolation	User A can never read user B's rows, on every /me/* route

TEST	RULE
Plans	A Free token calling a Pro endpoint gets 403 or reduced fields, never Pro data
Audit	Every recommendation and alert writes an audit row with the disclaimer shown
Load	The 15:45 scan plus brief generation finish within 15 minutes at the target user count

One product for every plan; your trading core kept apart

Free, Plus and Pro are not separate deployments: they share one codebase, one set of servers and one database, and each user's plan decides what they see. What does get separated is your personal trading bot from the public product.

PRIVATE · Trading core

existing droplet, static IP

Your Kite session and TOTP, paper and live orders, scan, models, training

↓ one-way publish: market state, decisions, DNA, scorecards

PUBLIC · Product stack

new, from Month 3–4

API, web and app, workers (brief, alerts, copilot, research), admin panel

Users, plans, billing, users' read-only broker links

Managed Postgres (Timescale) and Redis

+ a staging copy of both

WHY SPLIT	REASON
Security	Your Kite secrets never sit on the server that faces the public
Zerodha	API orders must come from the registered static IP, so trading stays on that droplet
Reliability	A traffic spike or product bug can't stop your stop-loss checks
Regulation	RA rules limit an analyst's own trading around recommendations; separate, logged systems make that auditable (confirm with the consultant)

How plans are separated inside one stack

Plan in the database

subscriptions and the plan catalogue hold what each user can use. The admin panel changes them without a deploy.

One gate in the API

api/deps.py decides fields, delay and rate limit per plan. The screens read the plan to lock or show features, but the API is the real gate.

Separate worker for heavy Pro work

Copilot, Research Lab and Monte Carlo jobs run on their own queue, so they can never slow the 15:45 scan, and that worker scales on its own.

Environments

ENVIRONMENT	DEPLOYS FROM	USED FOR
Local	Your machine, Docker Compose	Development and tests (Postgres on port 5433)
Staging new	Pull requests or a develop branch	Plan gating, billing in the payment gateway's test mode, three test accounts, your read-only Zerodha link
Production	main, after CI passes (as today)	Real users; trading core and product stack

Today a push to main deploys straight to production with no staging. That's fine for a personal tool, but not once people pay.

Testing with your own Zerodha

You keep testing everything with your own account. The plans only change what other people see.

WHAT TO TEST	HOW	WITH YOUR ZERODHA?
Your bot's paper and live trading	Unchanged: the trading core keeps your Kite session and the two-flag latch	✓ As today
Each plan's screens	Admin panel: View as Free / Plus / Pro, or change your own plan; three test accounts in staging	✓ No extra Zerodha accounts
Holdings features	Link your Zerodha read-only to your product account, exactly as a user would	✓ Your own data, no licence issue
Alerts, brief, copilot	Built from the intelligence your trading core produces	✓ For your own use
Outside testers	Paper trading only until the data licence and SEBI questions are answered	Needs legal check

Second Kite Connect app

The product stack likely needs its own Kite Connect app for users' read-only links, because each app has one redirect URL and yours points at the trading core.

Daily login

Kite tokens expire around 06:00 every day. The trading core logs in automatically; a read-only link in the product needs a daily re-login, like any user.

Static IP

As understood, the whitelist applies to placing orders, not reading holdings. Confirm with Zerodha support with the other Phase 0 questions.

When

Months 1–2

Single droplet; everything on your paper account

Month 3

Staging; accounts and roles; your owner account migrated

Month 4

Product stack split out; admin panel; billing in test mode; your read-only Zerodha link

Month 5

Private beta for you on real holdings; licensed data

Month 6

Outside users only if the launch gates pass

DAILY DATA

What runs each trading day, and who gets it

Times are IST, weekdays, extending workers/scheduler.py.

TIME	JOB	DATA USED	PRODUCES	FREE	PLUS	PRO
06:10	Broker login	Owner's Kite account	Session	Internal		
06:30	Pre-market fetch new	Global cues, FII/DII, event calendar, ASM/GSM lists	Market state inputs	Internal		
08:30	Market state new	Indices, VIX, breadth, sectors, regime change	market_state_daily	Regime label	+ trends	Full
08:45	Morning brief new	Market state, events, positions	Brief	Short, in app	Full, push	+ positions, all channels
09:15–15:30 · 60s	Quotes, stops, exits	Live quotes	SL HIT, exit states	End of day	Real time	Real time
09:15–15:30	Shock detector new	Price and volume jumps, filings	shock_events	—	—	Real time
Through the day	Announcements new	NSE filings	event_items	Badge next day	List	Real-time impact
15:40	End-of-day data	Candles, bhavcopy, delivery, deals, F&O	Features, factors, DNA	Internal		
15:45	Scan	Regime → sectors → setups → landmines → models → EV → portfolio → risk	decision_records	Internal		
16:00	Picks published	Decisions	Recommendations	5 large cap	Full list	+ EV ranking
16:15	What changed, signal scoring	Today vs yesterday	Changes, outcomes	—	Watchlist	All + portfolio
16:30	Backups, data-quality report	—	Off-site copy	Internal		
Evening	Position review, research jobs, failure labels	User data	Health, research, journal	—	Health score	Full
Monthly	Challenger training, drift report	Journal, predictions	Scorecards for approval	—	Summary	Model Lab

Sources

SOURCE	GIVES US	COST	STATUS	CAN USERS SEE IT?
Kite Connect	Candles, quotes, orders	₹500 a month	In use	Owner's own trading; plan a licensed vendor for users
NSE bhavcopy, delivery, deals, FII/DII, restriction lists, event calendar	Prices, flows, events	Free	In use	Derived scores and summaries, with sources
NSE announcement text	News brain	Free	Month 5	Yes, with the filing link
F&O bhavcopy, open interest	Positioning	Free	Month 4	Summary
Global cues	Overnight mood	Free to paid	Month 1	Summary
Licensed real-time vendor	Prices shown to users	Quote needed	Before launch	Within the licence
NSE XBRL results	Fundamentals	Free; weeks of work	12-month plan	Derived scores

Delivery channels

CHANNEL	FREE	PLUS	PRO	COST TO US
In app (web, PWA, Android, iOS)	✓	✓	✓	Hosting
Push	Brief only	✓	✓	Free
Email	Weekly	Daily	Daily	Small
Telegram	—	✓	✓	Free
WhatsApp	—	—	✓	Per conversation

GAP REGISTER

Every item, with its status, priority and month

A and B build on existing code; C is new; D is future. Priority follows the review: Core intelligence, then Differentiation, then Moat, with Platform alongside.

A1 · Market A · Built Regime and deployable capital Built Built · Effort — Free · Regime label Plus · Regime + deployable % Pro · Full rule view Today services/deployable.py maps NIFTY trend, volatility, breadth and VIX to six regimes and a deployable share (10% to 85%). Build Show it to users; add FII/DII to the rule table.	A2 · Portfolio A · Built Risk engine and capital ladder Built Built · Effort — Plus · Fixed 1% sizing Pro · Custom limits Today services/risk.py and limits.py: 1% risk per trade from stop distance, caps, cap-tier budget, cash floor, ₹10K–₹1Cr ladder, 15% drawdown halt. Build Per-user limits within the locked maximums.	A3 · Decision A · Built Landmine filter Built Built · Effort — Free · Applied to every pick Plus · Applied, with reasons Pro · Applied, with reasons Today services/avoid.py blocks entries near results and corporate actions, and for ASM, GSM or T2T stocks. Build Show the reason on screen (feeds F10).	A4 · Market A · Built NSE information feeds Built Built · Effort — Internal / platform Today services/market_feeds.py stores delivery %, deals, FII/DII, restriction lists and events daily, dated on fetch. Build Feed them into the market brain, DNA and the brief. Deals and FII/DII are stored but unused today.
A5 · Learning A · Built Calibration, walk-forward, Model Lab Built Built · Effort — Plus · Summary Pro · Full Today ml/calibration.py, purged 5-fold walk-forward, registry with rollback, Model Lab page. Build Base for the meta-model and the observatory.	A6 · Portfolio A · Built Exit policy Built Built · Effort — Free · SL HIT at end of day Plus · Real time Pro · Real time + trailing Today services/exit_policy.py: stop, target, trailing stop, half out at +5%, 30-day time stop. Build Base for exit intelligence (F13).	A7 · Learning A · Built Backtest replay Built Built · Effort — Internal / platform Today services/backtest.py and POST /backtest with the real cost model. Build Base for the Research lab.	A8 · Platform A · Built Broker, alerts, real-account tracking Built Built · Effort — Internal / platform Today Paper and live brokers behind a two-flag latch, Telegram alerts, holdings sync, real P&L report. Build Per-user alerts via P5; broker connections via P1.

F01 · Market B · Improve Market intelligence layer Core · Month 1 · Oct · Effort M Free · Risk-on / Neutral / Risk-off Plus · + NIFTY, BANKNIFTY, MIDCAP, SMALLCAP trends Pro · + global cues, liquidity Today market_context.py caches NIFTY, sector indices, VIX and breadth; deployable.py sets the regime. Build A morning market state: regime, index trends, global cues, FII/DII and liquidity, read by every model and screen.	F02 · Market B · Improve Sector brain Core · Month 2 · Nov · Effort M Plus · Sector ↑ → ↓ Pro · + industry, leaders, deterioration Today Sector relative-strength features, sector_map.py, sector cap. Build Daily sector and industry ranking on momentum, relative strength, breadth, money flow and volatility; leadership and deterioration.	F03 · Stock C · Build new Stock DNA Core · Month 4 · Jan · Effort M Plus · 4-trait summary Pro · Full profile, setups, regime, holding, failures Today 53 daily features and factor scores; no per-stock profile. Build Nine daily traits, plus learned best setups, best regime, typical holding and failure patterns from the journal.	F04 · Decision B · Improve Meta confidence system Core · Month 3 · Dec · Effort M Free · HIGH / MEDIUM / LOW Plus · 5 sub-scores + Expected R Pro · + calibration and model detail Today One calibrated probability, checked by calibration.py. Build Signal quality, ML probability, market context, risk quality and Expected R, combined by one calibrated meta-model.
F05 · Stock B · Improve Time-horizon probabilities Core · Month 4 · Jan · Effort M Pro · 3, 5, 10, 20-day odds + stop first Today One label: +8% before −4% in 15 bars. Build Horizon heads for +1%/3d, +3%/5d, +5%/10d, +8%/20d, and P(stop first).	F06 · Decision C · Build new Expected value engine Core · Month 2 · Nov · Effort S Plus · Expected R per pick Pro · Ranking by Expected R Today Research measured +0.22R for top-5% picks; costs.py has the cost model. Build $EV = P(\text{win}) \times \text{avg win} - P(\text{loss}) \times \text{avg loss} - \text{costs}$, in R; rank by EV.	F07 · Stock C · Build new Setup-specific models Core · Month 2 · Nov · Effort L Plus · Choose setups Pro · Per-setup probabilities Today One model per cap tier scores every stock. Build Rule detectors for breakout, pullback, momentum and mean reversion, each with its own model.	F08 · Stock C · Build new News classification Core · Month 5 · Feb · Effort L Free · Event badges Plus · Event list Pro · + materiality, sentiment, horizon Today Event dates stored and used to block entries; filings not read. Build Read NSE filings → company → event type → materiality → sentiment → horizon → meta-model input.

F09 · Stock

C · Build new

Event shock detector

Differentiation Month 5 · Feb · Effort M

Pro · Real-time shock alerts

Today Quotes polled every 60 s.
Build Price + volume jump + filing → review positions, re-rank watchlist, recalculate stops.

F10 · Decision

B · Improve

“Why not buy?” engine

Differentiation Month 1 · Oct · Effort S

Free · Top reason

Plus · Why and why not

Pro · + what would change it

Today Every signal stored with a rejection reason.
Build Evidence for and against, final state TRADE / WATCH / WAIT / AVOID, entry zone.

F11 · Decision

B · Improve

“What changed?” engine

Differentiation Month 3 · Dec · Effort S

Plus · Watchlist

Pro · All + portfolio

Today Confidence-decay alerts on open positions only.
Build Today vs yesterday for every stock, with point contributions and the main change.

F12 · Portfolio

B · Improve

Position intelligence

Differentiation Month 4 · Jan · Effort M

Depends on user input

Plus · Action state

Pro · + thesis tracking

Today My Holdings with daily model read, attention cards, stop alerts.
Build Entry thesis stored, health scored daily, HOLD / MONITOR / REDUCE / EXIT.

F13 · Portfolio

B · Improve

Exit intelligence

Differentiation Month 5 · Feb · Effort M

Free · SL HIT (end of day)

Plus · HOLD / REDUCE / TRAIL / EXIT

Pro · + exit reasons and score

Today Stop, target, trailing, half out, time stop.
Build Exit model with its own label and hysteresis.

F14 · Portfolio

C · Build new

Opportunity cost engine

Differentiation Month 6 · Mar · Effort S

Pro · Held vs new edge

Today Not built.
Build Remaining edge of holdings vs new setups; user decides.

F15 · Portfolio

B · Improve

Portfolio health

Differentiation Month 4 · Jan · Effort M

Depends on user input

Plus · Health score

Pro · Six scores + clusters

Today Capital page, limits, caps, deployable ceiling, calculator spec.
Build Diversification, risk, concentration, sector, exposure, liquidity → HEALTHY / WATCH / HIGH RISK.

F16 · Portfolio

C · Build new

Correlation engine

Core Month 2 · Nov · Effort S

Pro · Clusters, beta, cap concentration

Today Only a sector cap.
Build 60/120-day correlations, clusters, beta and cap concentration; warnings before entry.

F17 · Portfolio

B · Improve

Factor exposure

DifferentiationMonth 4 · Jan · Effort M

Pro · Price factors now; fundamental later

Todayfactors.py: trend and low volatility; others stubs; not wired.

BuildWire scores; momentum, volatility, size, liquidity exposure now.

F18 · Learning

B · Improve

Trading journal

CoreMonth 1 · Oct · Effort M

Free · Paper trades

Plus · Trades + outcomes

Pro · Full context + failure labels

TodaySignals with features, trades, predictions with outcomes.

BuildOne row per decision with context at entry, outcome in R and failure label. Must start early: it can't be back-filled.

F19 · User

D · Future

Personal trading twin

MoatMonths 7–12 · Effort L

Depends on user input

Pro · Personal ranking

TodayNot built.

BuildLearn holding period, setups, risk, sizes, accepted and rejected signals, mistakes → personal ranking.

F20 · User

D · Future

AI copilot

DifferentiationMonth 6 · Mar · Effort L

Depends on user input

Pro · Cited answers over your data

TodayNot built.

BuildRead-only tools over our data; “would reconsider if” answers. Owner beta in Month 6, users later.

F21 · User

C · Build new

Morning brief

DifferentiationMonth 2 · Nov · Effort S

Free · Short

Plus · Full

Pro · + your portfolio

Today16:00 Telegram summary only.

Build08:45: market, sectors, events, holdings needing attention, top setups by Expected R, avoid list.

F22 · Decision

B · Improve

Explainability

DifferentiationMonth 3 · Dec · Effort M

Free · Top 3 reasons

Plus · For and against

Pro · + feature contributions

TodayPlain-English reason per signal.

BuildEvidence lists from sub-scores, rules, events; SHAP for models.

F23 · Learning

B · Improve

Model observatory

DifferentiationMonth 6 · Mar · Effort M

Plus · Summary

Pro · Full lab

TodayModel Lab, registry with rollback, walk-forward, calibration.

BuildScorecards by regime; champion → challenger → walk-forward → paper → approval.

F24 · Decision

B · Improve

Data quality

CoreMonth 1 · Oct · Effort S

Free · Badge

Plus · Badge

Pro · Per-source detail

TodayFeed alerts, stale-list handling, spike guard.

BuildQuality score per decision; below threshold, no decision.

F25 · Decision C · Build new Uncertainty and model agreement Core · Month 3 · Dec · Effort S Free · “Low confidence: wait” Plus · + reasons Pro · + agreement count Today Not built. Build WAIT when models disagree (for example 4 of 6), data is poor, the regime is changing or an event is due.	F26 · Market B · Improve Regime change detector Core · Month 1 · Oct · Effort S Plus · Alert Pro · + re-rank and re-risk Today Regime computed daily. Build Two-day confirmation, then six actions: re-score, re-rank, review positions, recalculate risk, review stops, notify.	F27 · Learning B · Improve Strategy lab Differentiation · Month 6 · Mar · Effort M Free · Read-only catalogue Plus · Choose 1 Pro · All + stats Today Strategies tab, /strategies/performance. Build Best and worst regime, holding, drawdown, win rate, expectancy per strategy.	F28 · Learning B · Improve Research lab Differentiation · Month 6 · Mar · Effort L Depends on user input Pro · Backtest v1; walk-forward and Monte Carlo later Today Backtest engine and API; no screen. Build v1 screen in Month 6; v2 with walk-forward, Monte Carlo, what-if in Months 7–12.
F29 · Learning C · Build new Monte Carlo stress testing Moat · Months 7–12 · Effort S Pro · Drawdown range, survival Today Not built. Build Resample outcomes with streaks, slippage, gaps, volatility spikes, higher correlation.	F30 · Portfolio C · Build new What-if simulator Moat · Months 7–12 · Effort M Depends on user input Pro · NIFTY, VIX, cap, sector shocks Today Not built. Build NIFTY –5%, VIX × 2, small cap –10%, banking –8% → portfolio and stock impact.	F31 · Learning C · Build new Failure investigator Core · Month 5 · Feb · Effort M Pro · Loss labels and totals Today Post-exit watch is partial. Build Seven loss labels with detection rules; totals over the last 100 losses.	F32 · Learning D · Future Automated challenger loop Moat · Months 7–12 · Effort L Internal / platform Today Walk-forward, registry, shadows exist. Build Failure patterns → feature ideas → backtest → walk-forward → paper → human approval → challenger.

F33 · Portfolio

D · Future

Advanced portfolio optimisation

Moat Months 7–12 · Effort L

Depends on user input

Pro · Suggested weights to accept or edit

Today Calculator spec only.

Build Weights across candidates and holdings that maximise Expected R within risk, correlation and cap limits.

F34 · Stock

D · Future

Event-impact learning

Moat Months 7–12 · Effort M

Pro · Similar events and typical move

Today Not possible yet: event history starts in Month 5.

Build Measured moves after each classified event type; similar-event matching.

F35 · Market

C · Build new

Market memory (episode library)

Core Month 1 · Oct · Effort M

Free · Closest episode, one line

Plus · Episode card: what happened, what we did

Pro · Full memory + defence ablations

Today Five years of candles and a replay by 8-month window; nothing is split by market phase, and the regime limit isn't replayed.

Build Detect corrections, bear markets, shocks and weak phases by rule; replay the real system in each, with every defence; store results; match today to the closest episode.

F36 · Market

C · Build new

Bear mode and playbook

Core Month 2 · Nov · Effort M

Free · Bear-mode banner

Plus · + holdings flags, why fewer picks

Pro · + phase playbook, episode repeat on your portfol

Today Defences exist (regime limit, +0.10 bar, filters, drawdown halt) but users get no explanation.

Build Phase detection (early warning, decline, panic, recovery); playbook on selection and deployable levers only; bear-mode brief and holdings review.

F37 · Learning

B · Improve

Score report card

Core Month 2 · Nov · Effort M

Free · What each label has delivered

Plus · Report card by score band

Pro · + by regime and calibration detail

Today calibration.py buckets confidence against outcomes; evaluate_pending_signals scores every signal against target, stop and horizon.

Build Per band and regime: sample size, target-first rate, average R after costs, calibration gap; daily health check that flags bands falling 10+ points short.

F38 · Learning

B · Improve

Filter and factor scorecard

Core Month 2 · Nov · Effort M

Pro · Which filters and factors are working, by regime

Today Risk and execution rejections are stored and scored; filters inside the strategy leave no scoreable record; factors aren't wired.

Build Shadow scoring for every filtered candidate; per filter and regime: blocked count, what the blocked stocks did, verdict; daily factor spreads.

F39 · Learning

C · Build new

Lessons loop

Core Month 3 · Dec · Effort M

Plus · Lesson being applied

Pro · + full lessons report per episode

Today Not built.

Build When a weak phase ends: lessons report, playbook proposals, out-of-sample test on other episodes, normal-market check, approval, playbook version +1.

F40 · Decision

C · Build new

Trigger entry

Core Month 1 · Oct · Effort M

Plus · Trigger price on each pick

Pro · + trigger status alerts

Today Entries fill at the signal-day close; after-close regular orders are rejected live (roadmap blocker R2).

Build Test, then build: next session, buy only above the signal candle's high (+0.25%) on volume; unfilled triggers expire. Doubles as the after-close order method.

F41 · Portfolio C · Build new Structure stops, targets and time stop Core · Month 2 · Nov · Effort M Plus · Stop and target levels Pro · + breakeven and time-stop status Today Stop by ATR; fixed +8% / -4%; half out at +5%; 30-day time stop (locked spec). Build Test: stop below the swing low, targets by structure, breakeven at +1.5R, day-6 time stop, soft exit below the 5 EMA. Adopt only by owner decision if replay beats the locked spec.	F42 · Market B · Improve Big-investor deals Differentiation · Month 3 · Dec · Effort S Free · Top deals of the day Plus · All deals + your watchlist Pro · + alerts when a deal hits your holdings Today Bulk and block deals stored daily since 21 Sep 2026; read by nothing. Build Named investor deals by stock, buyer and seller, size vs volume; watchlist and holdings alerts; accumulation signal tested as a filter.	F43 · Portfolio C · Build new Community diversification benchmark Differentiation · Month 6 · Mar · Effort S Plus · Your score vs community Pro · + by capital band Today Not built; needs many users. Build Portfolio health score beside the anonymous average of all users (and users with similar capital).	F44 · Stock C · Build new Ready-made screens Differentiation · Month 4 · Jan · Effort S Free · 2 screens Plus · All ready screens Pro · + custom screens Today Scan results page with filters; no named screens. Build Named screens from the setup detectors: quality pullbacks, volume breakouts, leading-sector momentum, bear-mode safe list, big-investor buys.
F45 · Portfolio C · Build new Tax view Differentiation · Month 4 · Jan · Effort S Free · Estimated tax on paper trades Plus · On your trades Pro · + yearly tax report export Today Real P&L report is before tax. Build Estimated short-term capital gains tax on delivery trades (20% per current rules, confirmed with an accountant); after-tax P&L.	F46 · Market C · Build new Theme rotation Differentiation · Month 3 · Dec · Effort M Free · Top 3 trending themes Plus · Theme ranks + their stocks Pro · + theme momentum alerts and theme filter on se Today Sector indices and NSE index constituent parsing exist (c2646ba); no themes. Build Indian themes from NSE thematic indices and their constituent lists (e.g. Defence, CPSE/PSE, EV, Manufacturing, Digital); daily theme momentum, breadth and relative strength, ranked like sectors; stocks can belong to several themes.	F47 · Stock B · Improve Stock page Differentiation · Month 1 · Oct · Effort M Free · Header, decision label, daily chart, top reason Plus · + Expected R, setup, trigger, alerts, intraday cha Pro · + DNA, setups and odds, peers, overlays, full rec Today StockDetailModal shows card details and a daily closing-price chart with a lookback picker. Build A full stock page: header card with our decision, chart (1D-Max) with target, trigger and stop lines, EMAs, past signals; tabs for why / why not, DNA, setups and odds, events and deals, peers and themes, our record, financials later.	F48 · Platform C · Build new Upgrade and checkout page Platform · Month 4 · Jan · Effort M Internal / platform Today Not built. Build Plus and Pro with 1-, 3- (and 12-, if allowed) month billing, per-month equivalents, coupons with savings line, GST shown, refund terms, launch offer; coupon management in the admin panel.

F49 · Platform Sign-up and sign-in pop-up Platform Month 3 · Dec · Effort S Internal / platform Today Password and Google sign-in exist, limited by GOOGLE_ALLOWED_EMAILS. Build One pop-up from Sign up, Sign in or any locked feature: app-install QR panel; password or Google now; +91 phone OTP at go-live; consent to Terms, Privacy, RA and AI-use disclosures; then onboarding.	P1 · Platform Accounts, portfolios, broker connections Platform Month 3 · Dec · Effort L Internal / platform Today Sign-in works; trading tables have no owner. Build users → portfolios, watchlists, broker_connections; user_id and portfolio_id on owned tables; owner back-filled as user 1.	P2 · Platform Subscriptions and entitlements Platform Month 4 · Jan · Effort M Internal / platform Today Not built. Build Plans, payment gateway, trials, 6-month offer, gating in the API, pro-rata refunds.	P3 · Platform Compliance module Platform Month 4 · Jan · Effort M Internal / platform Today Not built. Build RA disclosures, MITC, AI-use disclosure, disclaimers everywhere, 5-year audit log, PaRRVA export.
P4 · Platform Licensed market data Platform Month 5 · Feb · Effort M Internal / platform Today Kite under the owner's account. Build NSE-authorised vendor for prices shown to users.	P6 · Platform Admin panel Platform Month 4 · Jan · Effort M Internal / platform Today users.is_superuser exists; sign-in limited by GOOGLE_ALLOWED_EMAILS in .env; Safety page has the kill and exits switches. Build Roles (owner, admin, support, user); users, subscriptions (including your own), view as any plan, plan catalogue, invites, approvals inbox, system and safety, compliance, revenue; admin audit log.	P7 · Platform Deployment split and staging Platform Month 4 · Jan · Effort M Internal / platform Today One droplet runs everything; push to main deploys production; no staging. Build Staging environment (Month 3); product stack separated from the private trading core, fed by a one-way publish (Month 4); separate Pro worker queue.	P5 · Platform Notification hub Platform Month 4 · Jan · Effort M Internal / platform Today Telegram to one chat. Build Per-user push, email, Telegram, WhatsApp; quiet hours; Free delayed.

Requirement chain

Feature → purpose → ML → data → backend → database → API → screen → tests → metric → priority, for the items filtered above.

ITEM	PURPOSE	ML / AI	DATA	BACKEND	DATABASE	API	SCREEN	TESTS	METRIC	PRIORITY
A1 Regime and deployable capital	Invest big only when the market is healthy.	Rule table by design.	NIFTY, VIX, breadth	services/deployable.py	system_state	GET /risk/limits	Capital page	Existing regime tests	Reproducible from stored data	Built
A2 Risk engine and capital ladder	Survival.	None	Positions, prices	services/risk.py, limits.py	existing	GET /risk/limits	Safety, Capital	test_risk.py, test_limits.py	Every block has a reason	Built
A3 Landmine filter	One bad gap undoes ten good trades.	None	Event calendar, restriction lists	services/avoid.py	existing	via signals	Signal cards	test_avoid_filter.py	Refusals explained	Built
A4 NSE information feeds	Information the chart doesn't contain.	Ablation before any model use	NSE archive files	services/market_feeds.py	daily_delivery, block_deals, institutional_flows	internal	—	test_market_feeds.py	Missed feed days	Built
A5 Calibration, walk-forward, Model Lab	Proves the probability is honest.	Existing	Signals, predictions	ml/calibration.py, ml/registry.py	ml_models, predictions	GET /ml/calibration	Model Lab	test_calibration.py, test_walk_forward.py	Calibration gap	Built
A6 Exit policy	Small losses, banked gains.	None	Positions, quotes	services/exit_policy.py	positions	GET /positions	Positions	test_trailing_stop.py	Exit reasons recorded	Built
A7 Backtest replay	Years of evidence in seconds.	—	Candles	services/backtest.py	in memory	POST /backtest	none yet	test_backtest.py	Matches paper accounting	Built
A8 Broker, alerts, real-account tracking	One code path for paper and live.	—	Kite	brokers/, notifications/	orders, positions, broker_fills	various	Holdings, Real report	existing suite	—	Built
F01 Market intelligence layer	Read the market before the stock.	Rules; context enters a model only if ablation shows it helps.	Index candles, FII/DII (have); global cues (new)	services/market_intel.py (new)	market_state_daily	GET /market/today	Home regime gauge, Market page	Reproducible state; regime mapping	Ready by 08:30 on 99% of trading days	Core Month 1 · Oct
F02 Sector brain	Money rotates; leaders lead.	Rules; the sector gate is tested on paper first.	Sector indices, delivery % (have); industry classification (new)	services/sectors.py (new)	sector_rank_daily	GET /market/sectors	Sector board; chip on every stock	Gate off gives identical scan	Top-3 sector picks vs rest, realised R	Core Month 2 · Nov
F03 Stock DNA	The identity of each stock.	Learned parts shown only with 20+ past setups, pulled toward the sector average.	Features, flows, events, journal	services/stock_dna.py (new)	stock_profile_daily	GET /stocks/{symbol}/dna	DNA tab	Traits match features; learned parts hidden below sample size	100% of universe daily	Core Month 4 · Jan

Item	Purpose	ML / AI	Data	Backend	Database	API	Screen	Tests	Metric	Priority
F24 Data quality	No data, no decision.	None.	Feed freshness	services/data_quality.py (new)	decision_reco_rds.data_quality	GET /system/data-quality	Freshness strip	Missing feed blocks decision	0 decisions on bad data	Core Month 1 · Oct
F25 Uncertainty and model agreement	Know when not to act.	Disagreement across models and heads.	F04, F24, F26	services/decision.py	decision_reco_rds.agreement	GET /stocks/{symbol}/decision	WAIT with reason	Each trigger forces WAIT	WAIT vs normal win rate	Core Month 3 · Dec
F26 Regime change detector	Adapt quickly.	Rules with hysteresis.	market_state_daily	services/market_intel.py	regime_events	GET /market/regime-events	Timeline	Replay finds known turns	False flips per quarter	Core Month 1 · Oct
F27 Strategy lab	Choose on evidence.	None.	Signals, trades	services/portfolio.py	existing	GET /strategies/performance	Strategies page	Stats match ledger	PaRRVA-verified before public	Differentiation Month 6 · Mar
F28 Research lab	Serious users test ideas.	Walk-forward built in.	Candles	services/backend.py + job queue	research_runs	POST /research/runs	Research Lab	Lab = CLI result	Runs per Pro user	Differentiation Month 6 · Mar
F29 Monte Carlo stress testing	Robustness.	Bootstrap.	Trade history	services/montecarlo.py (new)	research_runs	POST /research/runs	Fan chart	Analytic check	—	Moat Months 7–12
F30 What-if simulator	Scenario analysis.	Beta and correlation estimates.	Holdings, betas	services/whatif.py (new)	none	POST /me/whatif	What-if panel	Beta maths	Weekly use	Moat Months 7–12
F31 Failure investigator	Learn from mistakes.	Rules first.	Journal	services/failures.py (new)	trade_journal.failure_label	GET /me/failures	Failures report	100 hand-labelled losses	Unlabelled < 20%	Core Month 5 · Feb
F32 Automated challenger loop	Continuous improvement.	Experiment tracking.	Journal, failures	services/experiments.py (new)	experiments	internal	Model Lab	No path promotes without approval	Experiments per month	Moat Months 7–12
F33 Advanced portfolio optimisation	Allocation.	Constrained optimisation.	EV, correlations, limits	services/optimizer.py (new)	none	POST /me/portfolio/optimize	Allocation proposal	Never breaks a locked limit	Proposal R vs manual	Moat Months 7–12

ITEM	PURPOSE	ML / AI	DATA	BACKEND	DATABASE	API	SCREEN	TESTS	METRIC	PRIORITY
F34 Event-impact learning	Proprietary event intelligence.	Event-study statistics, then a model.	event_items + prices	services/news/impact.py (new)	event_impacts	GET /stocks/{symbol}/events	Impact panel	Out-of-sample event windows	Impact error	Moat Months 7–12
F35 Market memory (episode library)	Know what we did and lost last time, before it repeats.	Rules for detection; nearest-neighbour match on market state; no probabilities from so few episodes.	Index candles (have); NSE archive back-fill for 2018 and 2020	services/market_memory.py (new), services/backtest.py (episode mode)	market_episodes, episode_replays	GET /market/memory	Bear market memory page; episode card in the brief	Every 10%+ correction in stored history is detected; replay includes deployable %	Episodes covered; replay completeness	Core Month 1 · Oct
F36 Bear mode and playbook	Handle bear markets calmly and consistently.	Rules; playbooks proven out-of-sample on past episodes.	market_state_daily, episode results	services/bear_mode.py (new)	regime_events, playbook_runs	GET /market/today (bear block)	Banner, brief, holdings review	Locked risk rules unchanged in bear mode	Loss in weak phases vs NIFTY	Core Month 2 · Nov
F37 Score report card	Scores are accountable: “here is what 80+ has actually done”.	Calibration; refits only as approved challengers.	Scored signals (have)	ml/calibration.py (extend), services/scorecards.py (new)	score_band_stats	GET /scorecards/scores	Score report card	Numbers match the signal ledger; under-30 bands hidden	Bands under review; calibration gap	Core Month 2 · Nov
F38 Filter and factor scorecard	Keep what protects us, fix what costs us.	Counterfactual outcome scoring; factor spreads.	Signals incl. shadow signals, factor scores	services/scorecards.py (new), strategies/ml_swing.py (shadow records)	filter_scorecard_daily, factor_scorecard_daily	GET /scorecards/filters	Filter and factor scorecards in Model Lab	Every block leaves a scoreable record	Filters with a verdict; saved R	Core Month 2 · Nov
F39 Lessons loop	Each bear market makes the next one easier to handle.	Out-of-sample validation across episodes.	episode_replays, scorecards, journal	services/lessons.py (new)	lessons_reports, playbook_versions	GET /market/episodes/{id}/lessons	Lessons report; lesson shown on the bear-mode card	No playbook change without out-of-sample proof and approval	Our fall vs NIFTY's fall per episode	Core Month 3 · Dec
F40 Trigger entry	Buy only when the market confirms the setup.	None; rules tested in replay (T1).	Daily and live quotes	services/execution.py (order method), services/backtest.py (trigger mode)	orders.trigger_price	GET /signals (trigger)	Trigger price and status on signal cards	Replay: trigger vs close entry, same signals	R after costs; share of triggers filled	Core Month 1 · Oct
F41 Structure stops, targets and time stop	Exits matched to how each setup behaves.	Replay tests T3–T5.	Swing points, candles	services/exit_policy.py	positions (stop_source)	GET /me/positions/intel	Stop, target and time-stop status	Replay vs locked spec on the same trades	R per trade; share of full losses	Core Month 2 · Nov

ITEM	PURPOSE	ML / AI	DATA	BACKEND	DATABASE	API	SCREEN	TESTS	METRIC	PRIORITY
P1 Accounts, portfolios, broker connections	Required before any plan.	—	—	api/deps.py, db/models	users, portfolios, watchlists, broker_connections	/me/*	Sign-up, profile	User A never reads user B	Zero leaks in tests	Platform Month 3 · Dec
P2 Subscriptions and entitlements	Revenue.	—	—	services/billing.py (new)	subscriptions, entitlements, payments	/billing/*	Plans page	Free can't read Pro data	Payment success	Platform Month 4 · Jan
P3 Compliance module	Legal right to operate.	—	—	services/compliance.py (new)	audit_log, consents	internal	Disclosures	Every recommendation audited	100% audit	Platform Month 4 · Jan
P4 Licensed market data	Lawful data.	—	Vendor feed	services/data_vendor.py (new)	candles, quotes (source)	internal	—	Failover	Uptime	Platform Month 5 · Feb
P6 Admin panel	Run the product and test every plan yourself.	—	—	api/v1/endpoints/admin.py (new)	users.role, admin_actions, plan_catalog, invites	/admin/*	Admin panel	User token gets 403 on /admin/*; every action logged	Admin actions with a reason (100%)	Platform Month 4 · Jan
P7 Deployment split and staging	Security, reliability and a safe place to test plans and billing.	—	Published intelligence tables	infra/, .github/workflows/	Product database + trading-core database	—	—	Product stack holds no Kite trading secrets; staging deploys on every PR	Scan unaffected by product load	Platform Month 4 · Jan
P5 Notification hub	Right alert, right person.	—	—	notifications/	notification_prefs	/me/notifications	Alert settings	Free delayed; quiet hours	Delivery rate	Platform Month 4 · Jan

Two tracks, one gate a month

The intelligence track runs on the owner's paper book, so it needs no licence and doubles as the search for an edge. The product track builds what public users need. A month closes when its gate is green; a red gate moves the plan, not the gate.

Month 1 · Oct 2026

8 items

F01 Market intelligence layer

F10 "Why not buy?" engine

F18 Trading journal

F24 Data quality

F26 Regime change detector

F35 Market memory (episode library)

F40 Trigger entry

F47 Stock page

INTELLIGENCE TRACK

Find where the edge is lost (roadmap Track 3). Episode library and episode replay: what we did and lost in the 2021–22 and 2024–25 corrections. Market state, regime-change detector, decision records with why / why not, data-quality score. Start the journal and shadow scoring of every filtered candidate now, because neither can be back-filled. Replay tests T1–T10 from the swing research, starting with trigger entry (F40).

PRODUCT TRACK

Consultant engaged; data-vendor quotes; NISM exam booked.

Gate

A one-page answer on where the edge dies; a per-episode loss report for past corrections; 100% of decisions carry reasons and a data-quality score.

Month 2 · Nov 2026

9 items

F02 Sector brain

F06 Expected value engine

F07 Setup-specific models

F16 Correlation engine

F21 Morning brief

F36 Bear mode and playbook

F37 Score report card

F38 Filter and factor scorecard

F41 Structure stops, targets and time stop

INTELLIGENCE TRACK

Bear mode and playbook, with 2018 and 2020 episodes back-filled. Score report card and filter and factor scorecards (history replay first, live records as they fill). Setup playbook building blocks (swing points, support and resistance, Fibonacci, patterns); structure-exit tests (F41). Setup detectors and models (breakout, pullback, momentum), EV ranking, sector brain, correlation engine, morning brief v1 for the owner.

PRODUCT TRACK

RA application prepared; multi-user schema designed.

Gate

Setup + EV ranking running as a paper shadow beside the current strategy.

Month 3 · Dec 2026

9 items

F04 Meta confidence system

F11 "What changed?" engine

F22 Explainability

F25 Uncertainty and model agreement

F39 Lessons loop

F42 Big-investor deals

F46 Theme rotation

F49 Sign-up and sign-in pop-up

P1 Accounts, portfolios, broker connections

INTELLIGENCE TRACK

Meta confidence model, uncertainty and model agreement, what changed, explainability, lessons loop. Mean-reversion model if it passes.

PRODUCT TRACK

Accounts, portfolios, watchlists, broker connections; owner migrated as user 1.

Gate

Calibration gap ≤ 10 points; isolation tests green.

Month 4 · Jan 2027

13 items

F03 Stock DNA F05 Time-horizon probabilities

F12 Position intelligence F15 Portfolio health

F17 Factor exposure F44 Ready-made screens

F45 Tax view F48 Upgrade and checkout page

P2 Subscriptions and entitlements P3 Compliance module

P6 Admin panel P7 Deployment split and staging

P5 Notification hub

INTELLIGENCE TRACK

Stock DNA traits, time-horizon heads, portfolio health, factor exposure, position intelligence.

PRODUCT TRACK

Billing and plan gating, compliance module, notification hub, all in test mode.

Gate

DNA for the whole universe; failing horizon heads hidden; Free cannot read Pro data.

Month 5 · Feb 2027

5 items

F08 News classification F09 Event shock detector

F13 Exit intelligence F31 Failure investigator

P4 Licensed market data

INTELLIGENCE TRACK

News classification, event-shock detector, exit model, failure investigator.

PRODUCT TRACK

Licensed data contract and integration; private beta with a few testers (no public calls).

Gate

News $\geq 85\%$ on labelled filings; exit model beats current exits in replay.

Month 6 · Mar 2027

6 items

F14 Opportunity cost engine F20 AI copilot

F23 Model observatory F27 Strategy lab F28 Research lab

F43 Community diversification benchmark

INTELLIGENCE TRACK

Opportunity cost, model observatory with champion / challenger, strategy lab, Research lab v1, copilot for the owner.

PRODUCT TRACK

Launch decision: public Free and Plus only if RA is granted, data is licensed and paper evidence shows an edge.

Gate

3 months of paper: EV ranking beats confidence ranking after costs.

Months assume the current pace of work on this repository and are planning estimates. Public launch happens only when SEBI registration is granted, the data licence is signed, and paper evidence shows an edge. Otherwise Month 6 ends in a private beta.

The moat is data nobody else has

Features can be copied. What can't be copied is years of our own decisions, outcomes, classified events and user choices, each recorded point-in-time. Everything in Months 7–12 turns that record into an advantage.

Proprietary data that compounds

Decision journal

Every decision with its full context and outcome. Feeds DNA, failure patterns and every model retrain.

Event-impact library

Classified announcements with measured moves afterwards. It can only start the day recording starts.

Failure-pattern library

Why losses happened, by setup, regime and stock. It turns losses into model improvements.

Verified track record

PaRRVA-verified performance that grows each month and is required before any public claim.

User decisions

Accepted and skipped signals (with consent) that train each user's trading twin.

Model lineage

Every champion, challenger and promotion decision with its evidence.

Months 7–12 build list

F19 · Personal trading twin

D · Future

Learn holding period, setups, risk, sizes, accepted and rejected signals, mistakes → personal ranking.

Preference model; needs 50+ decisions.

F29 · Monte Carlo stress testing

C · Build new

Resample outcomes with streaks, slippage, gaps, volatility spikes, higher correlation.

Bootstrap.

F30 · What-if simulator

C · Build new

NIFTY −5%, VIX × 2, small cap −10%, banking −8% → portfolio and stock impact.

Beta and correlation estimates.

F32 · Automated challenger loop

D · Future

Failure patterns → feature ideas → backtest → walk-forward → paper → human approval → challenger.

Experiment tracking.

F33 · Advanced portfolio optimisation

D · Future

Weights across candidates and holdings that maximise Expected R within risk, correlation and cap limits.

Constrained optimisation.

F34 · Event-impact learning

D · Future

Measured moves after each classified event type; similar-event matching.

Event-study statistics, then a model.

F20 · AI copilot for users

Opens to Pro users once the legal opinion clears it.

F28 · Research lab v2

Walk-forward, Monte Carlo and what-if as queued jobs.

XBRL · Fundamentals from NSE XBRL

Unlocks value, quality and growth factors and earnings traits in DNA.

DECISIONS

What I need from you

- ☐ **Freeze this architecture**, including the four corrections to the review: one meta-model, news impact only after history, User Brain feeding ranking not DNA, gates before public launch.
- ☐ **Adopt the priority order**: Core intelligence → Differentiation → Moat, with the product track in parallel.
- ☐ **Go multi-user?** Selling plans reverses roadmap guardrail R13 and commits us to SEBI registration, licensed data and billing.
- ☐ **Budget for Month 1**: consultant, data licence quotes, NISM exam.
- ☐ **Language model**: the Claude API for the brief's wording, news classification and the copilot.

After you approve

1

You mark changes on this document

2

An implementation plan for Month 1, one task per module, tests first

3

You choose how it's executed

4

Build behind flags; byte-identical scan when off

5

Check the month's gate, then plan the next month

A planning document. Nothing here is a projection or guarantee of returns. Backtests, paper results and calibration describe past behaviour, not future capital. Mock-up values are illustrative.